



Drone cybersecurity issues, solutions, trend insights and future perspectives: a survey

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Abstract

This paper presented an exhaustive survey on the security and privacy issues of drones. These security concerns were thoroughly dissected, particularly the aspect of cybersecurity, which was classified into nine levels. These levels include emerging issues, communication-based attacks, sensors, hardware, hardware-based attacks, software attacks, and physical attacks on the drone itself. Furthermore, we discussed the other non-cybersecurity challenges of drones, such as terrorism, mid-air collisions, illegal surveillance, smuggling, electronic snooping, and reconnaissance, alongside proffering possible solutions. Many of the discovered aspects of drone cybersecurity issues were then quantitatively analyzed using a multi-criteria decision-making problem-solving technique. The questionnaire responses from the general public, experts, and stakeholders in the aviation industry were analyzed. The findings revealed variations in cyber-attack techniques such as distributed denial-of-service (DDoS), denial-of-service (DoS), hacking, jamming, spoofing, electronic snooping, eavesdropping, advanced persistent threat (APT), reconnaissance, hijacking, man-in-the-middle attack, and so on. However, the majority of the participants in the survey, which constitute 70%, were unaware of the existing drone cybersecurity challenges. The remaining 30% were aware of the current drone security issues. Meanwhile, both parties are looking for an immediate solution that will fully provide an atmosphere of prospects in the drone industry. Following that, we presented our experience with drone security and privacy, as well as potential future research directions. This paper is unique in that it discusses the various types of drone cyber-attacks and non-cyber-attack scenarios that threaten the socio-economic system, aviation industry, national security, as well as public security and privacy concerns. It also offers solutions to the cyber-attack and non-cyber-attack cases that have been investigated. As a result, the findings of this study could be used to create, develop, and implement more secure cloud systems to safeguard drones from cyber and non-cyber-attacks.

Keywords Drone and cybersecurity · Drone cybersecurity challenges · Trending insight of cyber-attack · Internet of drone security · National security · Security and privacy and cyberspace future

1 Introduction

With an increasing number of drones flying daily, there is a growing need for coordination in safety, secure communication, and information sharing among these devices [96, 142, 311]. Interestingly, drones are a viable and

competitive option with landed vehicles for various applications like search and rescue and environmental monitoring. For instance, the use of drones to monitor industrial activities in the oil and gas industry in various communities, such as the United Arab Emirate (UAE), has increased in recent time [13, 28, 131, 256, 274, 310, 311]. Protecting drones through the provision of secure flight and data communications is necessary given the significant usefulness of drones to ensure mission success and guide against unprecedented accidents or attacks. While the hardware and software architecture of drones were designed with security in mind, the technology was not without flaws or dangers of its own. Thus, in recent time, the drone system has witnessed a rising number of cyber-attacks since 2007 [88, 126]. In addition, drones have proliferated without

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strict control, posing security risks, particularly in developed nations like the USA and Europe [97, 148, 326]. Drone payload and mission control hardware and software architecture are examples of critical infrastructure that could be attacked by adversaries [190, 320]. Drone systems' radio links [38, 170] carry information like data requests via control signals, cellular user equipment (UE), and global positioning system (GPS) signals for drone navigation, and thus, could be the target of cyber-attacks.

Intruders have compromised the drone's data link and other vulnerable areas [22, 277]. Information is sent to and received from the drone via radio frequency (RF) transmission via data links, which is especially useful during flight missions [172]. These transmissions can include location, the location of the controller, distance to the ground control system (GCS), distance and location to the target, payload information, altitude, airspeed, remaining flight time, and various other parameters. The data link can also transmit live video between the drone back and the GCS, so the controller and the ground crew can watch what the drone camera is seeing. Adversaries, for example, have intercepted communications and stolen data disseminated and requested by drones due to vulnerabilities [22, 277]. Likewise, due to drones' vulnerabilities, attackers have physically operated drones by exploiting their control signals [146, 270, 287]. Due to the fact that both data and control signals are transmitted via radio links, protecting these open networks is crucial to the safety of the drone system as a whole.

Studies have shown that a plethora of cybercrimes have occurred, such as intercepting unencrypted data, images, and video from links or spoofing the drone's network and communication protocols [146, 219]. The hardware and software vulnerability and complexity of drones raises possible security and privacy concerns since the majority of existing commercial drones lack security protocols such as intrusion detection systems (IDS). Over the years, a lack of security measures has affected drone success in missions [44, 148, 258, 303]. Furthermore, the software and hardware design complexity of drones raises possible security and privacy concerns.

The current solution to drone cyber-attacks includes the use of several cryptographic hash algorithms that have been proposed over the last 30 years. The previous and existing algorithms, such as Secure Hash Algorithm-1 (SHA-1), Message Digest 5 (MD5), Whirlpool, and RACE Integrity Primitives Evaluation Message Digest (RIPEMD), are vulnerable to length extension and collision forms of attacks. However, more recent solutions, for example, truncated SHA-3, SHA-2, BLAKE3, and BLAKE2, can handle these types of cyber-attacks. Moreover, the extensively adopted public key cryptographic schemes and algorithms typical of the Public Key

Infrastructure (PKI) framework enable secure data sharing via digital certificates that bind an entity like an object to its public key [65, 123, 184]. By applying PKI, the legitimacy of the Internet of Drones (IoD) network entities can be verified, and information can be securely shared. Furthermore, Elliptic-Curve Cryptography (ECC) has been employed to provide a similar level of security as the Rivest–Shamir–Adleman (RSA) cipher block key while using significantly smaller key sizes.

The RSA algorithm is a prominent finite field exponentiation over integers, including prime numbers. The approach uses sufficiently large integers, making it challenging to solve. In the RSA algorithm, there are two types of keys: private and public. The use of Mobile Edge Computing (MEC), Machine Learning (ML), physically unclonable functions (PUF), block chain, and Elliptic-Curve Cryptography (ECC) is now trending and can be explored to technically address drone security. Since numerous new use cases for drones have surfaced, there is now a pressing need to look into security concerns in order to ensure the continued success of drone missions. Therefore, a thorough and comprehensive strategy is required to guarantee the success of drone flight missions. The challenges of addressing them have been enormous for business, academia, and governments. As a result, the authors of this study analyze the cybersecurity and cyber-attack problems plaguing drones in order to propose substantial solutions and call attention to the need for further study in this area. Meanwhile, the main hardware components of the drone system and ground control station are depicted in supplementary Figure 1.

A drone is a system made up of six major submodules, which are (i) the flight computer, (ii) the gyro, (iii) payload control, (iv) the communication subsystem, (v) cameras and (vi) sensors, all of which work together to create a highly valuable observation platform. Supplementary Fig. 1 depicts a schematic view of each submodule. All six main submodules and other hardware components of the drone system can be subjected to malicious threats and attacks.

Aside from that, the burgeoning use of drones has been influenced by innovations in camera and sensor technology. A variety of communication mechanisms, including radio modems, microwave links, satellite communication, and others, ensure a continuous link between the drone and the ground base station. Radar, such as Doppler radar, is used as a measuring tool for the speed of the drone. Current drone technology offers feasible technical solutions for flight control, airframes, communications, and base stations. There are two elements that limit the flexibility of the drone system: human intervention and mission flexibility. However, the aspect of human control of the drone from the ground station is still required, as flight control

computers do not provide additional support beyond basic flight plan definition and operation. Most of the time, payload is remotely operated with no or very little automation support. Economic efficiency necessitates the ability of the same drone to operate in multiple application domains. This translates into stronger requirements for the mission and payload management subsystems, with increased levels of flexibility, automation and security.

With its vast number of airspace users, the drone industry, also known as unmanned aerial vehicles (UAVs) or remotely piloted aircraft (RPA) or the Internet of Drones (IoD), has seen a rapid development in the twenty-first century. However, there are difficult security and privacy issues that must be addressed for civilian and military application success. Cellular technologies such as Advanced—5G, 4G, GPRS, UMTS/WCDMA, EDGE, UMTS/HSPA, LTE, and LTE-A do not have the same frequency. This paper investigates the security implications of drone technology's rapid growth, focusing on emerging and current vulnerabilities as well as future trends. The likelihood and potential consequences of security threats increase as drones become more popular and sophisticated in national development. In fact, sales of piloted or autonomous drones are expected to exceed \$24 billion by the year 2030.

2 Motivation

Drones have gained popularity in real-world applications just like social networks [196]. But due to their reliance on wireless connections, drones are susceptible to various cyber-based attacks. Commercial and non-commercial losses are just two examples of the harm that can result from these kinds of attacks. This is likely due to a misunderstanding of the strategies hackers use to take control of a drone and illegally access, cause damage, or even cause the drone to crash. It is possible for criminals to hijack drones. Hence, they must be detected and prevented from causing harm. Importantly, drones, by concept, lack a failproof end-to-end security solution in order to reduce design and manufacturing costs. Factually, to keep costs down during development and production, drones are conceptually designed without a comprehensive security system. Common security flaws in commercial drones include vulnerabilities to firmware manipulation, a lack of encryption for both static and transmitted data to the ground controller. Security and privacy issues are also threats to the Internet of Drones (IoD) infrastructure because of the heterogeneity of connected devices. They are confronted by intrusion attempts, which might jeopardise the security and privacy of the acquired data during a mission. Drones are now more susceptible to a variety of

cyber and non-cyber threats. These attacks can have far-reaching consequences, including financial and non-financial damages.

In reality, drones can be hacked for malicious reasons. It is critical to detect attacks on drones and prevent them from causing more harm. Aside from a comprehensive review of previous and current challenges, this work explores emerging areas of drone security threats including but not limited to terrorist attacks, mid-air collisions, illegal surveillance, smuggling, electronic snooping, reconnaissance, and so on. Therefore, part of the recommendation includes the use of blockchain for analysis to protect the privacy of drone big data [168] and artificial intelligence models to address or mitigate cyber-attacks on drones.

2.1 Paper organization

In this paper, we explore some of the most pressing issues and difficulties associated with protecting drones from cyber and physical attacks. Key components of it are marked off as follows: Sect. 1 covers the introduction; Sect. 2 presents the motivation as well as the paper structure; Sect. 3 covers the study's contributions to knowledge, including the solutions to the drone security issues. Cybersecurity issues, unanswered questions, current and emerging trends are also covered. Section 4 discusses communication issues. Section 5 discusses sensor issues. Section 6 discusses hardware-related issues. Section 7 discusses hardware-based attacks. Section 8 discusses software issues. Section 9 discusses software-based issues. Section 10 discusses physical attack issues. Section 11 discusses emerging security issues. The regulatory concerns related to drones are discussed in Sect. 12. Section 13 discusses existing works that combine previous and current studies, each with their own subsections. Section 14 discusses methodology, which introduces search keywords, the data collection process, and the main analysis used for the data. It also provides a flowchart that describes in detail how the survey was conducted methodically. Data from the survey, along with analysis and interpretation, are presented in Sect. 15. Section 16 is the discussion of the results. Section 17 is about future directions. Section 18 contains the recommendations. Finally, Sect. 19 draws the conclusion.

3 Contributions

This study contributes to knowledge in nine major areas of drone security issues, which are classified and summarized as follows

- (i) It identified nine areas of security issues and categorized them as (1) emerging security issues, (2) communication issues, (3) communication-based attacks, (4) sensors issues, (5) hardware issues, and (6) hardware-based attacks. Others are (7) software-based attack issues, (8) software issues and (9) physical attack issues. These nine important categories were divided into separate issues and thoroughly examined as shown in Fig. 2.
- (ii) Mobile ad hoc networks (MANET), solutions to emerging security issues, physical attack issues, technical issues (communications, hardware, software, and sensors), privacy concerns, and regulatory problems with drones are just a few of the topics addressed in this work that were not covered in the aforementioned surveys.
- (iii) We provide a deeper insight of drone insecurity by thoroughly examining their security requirements and objectives. These requirements and goals include primitives such as, availability, authentication, authorization, confidentiality, integrity, privacy, and non-repudiation.
- (iv) The study provides a detailed overview about the threats and vulnerabilities targeting drones for each security level.
- (v) Likewise, we explored the aspect of cyber espionage which has emerged as a significant threat to drone development in recent time.
- (vi) Then, we reviewed the attacks and their existing countermeasures, identified the gaps in the literature, and explored privacy concerns in the era of drones before moving on to analysis, discussion, future perspectives, and making recommendations.
- (vii) Finally, we discussed open research challenges, current drone standards limitations, and potential future directions, recommendations, regulatory, detection, and prevention strategies.

The main contributions of this research as well as the distinctiveness of its security issues are summarized in supplementary Fig. 2.

This paper focuses on several security measures that can be taken to protect drone from cyber crooks. Meanwhile, the security issues in Fig. 2 are further discussed in the succeeding sections.

4 Communication issues

Many kinds of drones have two communication units, that is, a receiver and a transmitter. The receiver unit is the drone, and the transmitter unit is typically a ground-based

controlling system that allows the user to send commands (i.e., operations, signals, messages, instructions, directions, data, information, and so on) to the drone. However, in order to offer an appropriate level of data confidentiality, integrity, and availability, secure communication is a vital necessity in many instances. Drone communications are gaining popularity due to their numerous applications and benefits, such as low cost, high mobility, and on-demand deployment for rescue operations, disaster relief, and delivery missions. However, there is a problem with physical-layer security in a network. In the presence of a prospective eavesdropper and both on the ground, a drone's communication system relays secret information to a legitimate receiver [255]. The system's secrecy rate must be maximized by optimizing the drone's trajectory and transmitting power across a finite horizon together.

Likewise, many drone procedures are insecure and constitute a major hazard. Implementing a multi-drone network at the communication level must take into account potential security risks based on the network topology chosen. Taking potential security risks is significant because communication is an critical component of the drone system. Standardized drone protocols that enable reliable and secure communication should be created. The majority of present drone communication protocols are unencrypted or have weak cryptographic capabilities, making it possible for attackers to breach communication channels. Furthermore, currently available security solutions to safeguard civilian drones from criminal operators are confined to single drone systems [253]. Communication is the crucial component of the drone system for control during flight as well as data transmission. The majority of drones communicate with the ground control system (GCS) via wireless communication. There are numerous communication-level risks, vulnerabilities, threats, and attacks against drones today that jeopardize their secrecy, integrity, authenticity, and availability. There is an urgent need to know the type of wireless communication technology that guarantees a high level of security for drones. Therefore, there is a need to develop countermeasures for multiple drone attack scenarios.

- (i) *Data Transfer Speed is Slow* One of the drawbacks of using drone technology in precision agriculture is the slow data transmission speed, which some experts believe could take up to a week. The operation of the drone would be a waste if the time required for data transfer results in a farmer's inefficiency and causes harm to fertilizers, crops, or pesticides. So, if data transfer speeds are slow, pain and damage could happen during this time, which would make all of the work pointless.

- (ii) *Communication vulnerabilities* Communication vulnerabilities and threats can be classified based on the layers of transmission as follows: Vulnerabilities and threats at the physical and medium access control layers of the drone's 2-GCS wireless communication network create possible weaknesses. The operational frequency of a wireless communication system is increased from the low-frequency band to the millimeter-wave band due to the unavoidable tension of frequency spectrum resources [163]. Similarly, recent advancements in fifth-generation (5G) communications have increased interest in millimeter-wave bandpass filters (BPFs) [302]. The study by Hooper et al. [120] demonstrated three separate attacks against commercial Wi-Fi-based drones. The three types of attacks are buffer overflow, denial of service (DoS), and address resolution protocol (ARP) cache poisoning. An ARP spoofing attack, also known as ARP poisoning, is a man-in-the-middle (MitM) exploit that allows attackers to eavesdrop on network communication by sending a fake ARP message into a TCP/IP network as a way of intercepting packets. Their findings highlight serious security flaws in drone 2-GCS wireless connections. The sort of wireless communication technology to use is determined by the mission requirements' specifications, for example, transmission range, operating frequency, category, etc. However, because there is a need to evaluate the security risks of each form of wireless communication system, this choice does not ensure the flight's success. Hence, there is a need to resolve the issue of finding a wireless communication system for each application domain that offers a high level of security for drones.
- (iii) *Vulnerabilities and threats at the network layer* The drone network is made up of ad hoc networks known as flying ad hoc networks (FANETs). FANETs, or "flying ad hoc networks," are a novel type of mobile ad hoc network (MANET) in which the nodes are drones. Single-drone systems cannot become FANETs under this description, which is only acceptable for multi-drone systems. However, not every multi-drone system is part of a FANET. These FANETs have a complex architecture and pose a serious threat. Drone-assisted public safety networks pose a variety of security risks, according to previous research [113]. It demonstrates that as the complexity of a drone network grows, so does its vulnerability to attacks. Sensor inputs and communication modules are the major targets of these attacks.

A potential DoS attack could be caused by drone communication risks, such as intercepting or interrupting the communication link between the flight controller and the ground control station. Furthermore, given the specific properties of FANETs, such as latency and processing capacity for data routing, cryptographic methods for FANETs must take these aspects into account [61]. By transmitting malicious communications directly through the GCS or indirectly through the drones, an attacker can disrupt the drone network. The GCS is continually endangered by adversaries, either in a centralized or decentralized structure. The GCS is a single point of failure in both topologies, and the security of the entire drone network is dependent on its security. Despite the implementation of security safeguards for the GCS, an attacker can still disrupt the flight operation by compromising the flying drones. It should be noted that in some cases, even if one or more drones are compromised, the flight mission can still be judged successful. In this situation, and depending on the civilian use, the operator will only need a little amount of money.

- (iv) *Vulnerabilities and threats in the transport layer* If not adequately protected, drone communication protocols have weaknesses that can lead to a variety of attacks. Regardless of their communication capabilities, they must meet basic security standards, including confidentiality, integrity, availability, and authenticity. MAVLink, one of the most well-known drone communication protocols, has recently been discovered to be vulnerable to internet control message protocol (ICMP) flooding and packet injection attacks, according to recent research by Kwon et al. [153]. A Ping flood DDoS exploit, also known as an Internet Control Message Protocol (ICMP) flood DDoS attack, is a typical denial-of-service (DoS) attack in which an adversary attempts to overwhelm a targeted system with ICMP echo-requests (pings). ICMP Flood DDoS attacks are dangerous because they can flood the targeted device's network connections with fraudulent traffic, thereby hindering the passage of legitimate requests. This scenario increases the risk of DoS, or DDoS in the case of a more coordinated attack. The fact that in the past, attackers would fake a phony IP address to hide the sending device makes this volumetric attack vector considerably riskier. Today's sophisticated botnet attacks, particularly IoT-based bots, do not really bother concealing the bot's IP address. Instead, they flood the target server with

a large network of unspoofable bots. Attack points for different drone network architectures are presented in supplementary Table 1.

- (v) *Physical and MAC layer attacks* Given the considerable differences between ordinary wireless networks and aerial networks and ordinary wireless networks, the best wireless technology for drones must be chosen [112]. The summary of the primary features and security concerns of wireless communication technologies in drone devices is shown in supplementary Table 2.

In supplementary Table 2, we categorize each wireless communication method at the physical and MAC layers, as well as its unique features and specific security challenges, in this context. Although studies on the security of each wireless communication system can be found separately. The industrial control system has become an open structural system as information technology advances because it is always connected to the Internet and the Internet of Things [169]. However, the industrial control system's ability to operate consistently is severely hampered by this open structure, which exposes the system to security risks and makes it vulnerable to external invasion attacks.

Network Layer Attacks: These are cyber-attacks that target the network layer. DoS, eavesdropping, man-in-the-middle, replay, forgery, and other forms of attacks on the FANETs are examples of assaults on the network layer of drone communications.

- (a) *Eavesdropping attacks* By gathering data such as live video feeds, sensor readings, and GPS data supplied by the drone to the GCS, an attacker can undertake an eavesdropping assault over the UAV-2-GCS communication channel. This is due to the fact that most drones do not encrypt wireless transmission in order to improve communication performance [88, 127, 283]. The attacker can listen in on telemetry feeds and GCS commands, among other things. As a result, by acquiring sensitive information such as sensor readings and GPS data, the attackers can compromise the secrecy of communication and data.
- (b) *DoS attacks* By initiating a DoS assault, an opponent can compromise a drone system. In this situation, the attacker can send repeated requests to flood the flying drone's network card with random traffic, overloading its resources and breaking its availability. The impact of such assaults on drones can lead to a significant rise in network latency and a reduction in the quality of video streaming applications for users [181]. Sending numerous packets to the GCS within a certain range to deactivate the control signal is another approach to launching a DoS attack. When

the signal is lost, the drone enters a “lost connection” state, which causes the data link to malfunction. As a result, the operator is no longer able to operate. Ultimately, implying the operator can no longer send or receive data signals to the flight controller, causing the communication link to be disrupted and the drone to lose control. Lang et al. [154] used botnets to mimic a Distributed DoS (DDoS) assault on drones in a study. By flooding network traffic with User Datagram Protocol (UDP) packets, a DDoS attack was simulated. This type of simulation shows how real-world DDoS attacks on drones can be carried out. Additionally, de-authentication assaults may prevent the operator from commanding the drone. De-authentication is a denial-of-service attack that involves sending de-authentication packets to drones in order to interrupt drone-2-GCS connectivity. Hence, the enemy disables drone-2-GCS connection, causing the drones to lose network connectivity. Skyjack is a good illustration of this type of attack [6].

- (c) *Man-in-the-Middle attacks* The attacker controls the drone-2-GCS wireless channel and modifies the innocuous packets with malicious information in this well-known attack [285]. As a result, the opponent can function as a conduit between the drone and the GCS, potentially jeopardizing bidirectional drone-2-GCS communication. A video replay attack is a type of man-in-the-middle attack in which the attacker deceives the operator by sending fraudulent live feed data [88].
- (d) *Forgery attacks* By sending a fraudulent request to unauthenticated drones, the adversary can jeopardize drone communication integrity [327]. In this attack, the adversary impersonates a genuine request to construct a malicious request that breaks the drone-2-GCS connection.
- (e) *Replay attacks* The attacker can use an eavesdropping attack [316] to intercept many requests on drone networks, then replay valid data to the drones. In this instance, the drones may get duplicate data, and if no replay protection is in place, the drones may be unable to discriminate between legitimate and fraudulent requests [327].
- (f) *Attacks on FANETs routing* In MANETs routing protocols, different passive and active cyber-attacks can occur, such as manipulating network traffic, introducing rogue nodes, or disrupting routing functions [175]. Since FANETs are a subtype of MANETs, most existing attacks targeting routing protocols on MANETs can be transferred to routing protocols on FANETs. To demonstrate these attacks, we divide them into three groups based on how they

route data [176]: (i) The wormhole [121], blackhole [281], sybil [77], and sleep deprivation [223] are all route-finding attacks that target traffic control. (ii) Route maintenance attacks: these attacks attempt to tamper with routing control packets. Examples of route maintenance attacks are flooding [134] and Byzantine attacks [26]. (iii) Data forwarding attacks: these are attacks that have an impact on the payload traffic, such as real-time video traffic [122].

- (g) *Transport layer attacks* They are a type of cyber-attack. Attacks against the transport layer of UAV communication can be classified according to the protocols used by the transport layer.
- (h) *UranusLink Protocol Attacks* UranusLink solely provides integrity protection via the checksum field in the messages, according to the design and implementation of UranusLink for application in a real-world scenario [147]. An enemy with the ability to collect the exchanged packets, on the other hand, could exploit this flaw and reveal mission-critical information [140]. Thus, the data transferred between the drone and GCS will be collected. In addition, the attacker is capable of spoofing system IDs. The work by Highnam et al. [117] described a realistic scenario of many drones operating under the MAVLink protocol being compromised. By capturing a flight mission's system ID and spoofing MAVLink packets, the considered specimen threat scenario exhibits an attacker's capacity to carry out a stealthy assault.
- (i) *Attacks on the MAVLink Protocol* MAVLink attacks are classified into four categories by Laurito et al. [155] based on how data are compromised: interception, alteration, interruption, and fabrication assaults. Because the MAVLink protocol lacks authentication and encryption, the attacker can eavesdrop on communication traffic and hence, acquire data sent between the GCS and the drones. Different security mechanisms to maintain secrecy, authentication, availability, and data integrity in drone communications have been proposed in the literature as countermeasures for communication-based attacks. The highlights of drone network and transport layer communication security issues, existing limitations and countermeasures are presented in supplementary Table 3.

Thus, supplementary Table 3 outlines the security challenges related to drone network and transport layer communication, as well as available workarounds and restrictions. Using threats based on communication, the enemy can cut off the link and put the flight operation at risk. This can happen at several levels. Specific

countermeasures have been established in the literature to ensure the confidentiality, integrity, and availability of the shared data. Building IDS solutions, utilizing validated encryption to mitigate eavesdropping attacks, providing a multi-layer security architecture, and implementing a secure-based routing mechanism are all examples of countermeasures. Nevertheless, it is important to note that the above-mentioned defenses for drone communication-based cyber-attacks have various limits and flaws. Building IDS solutions to avoid DoS attacks, for instance, has an impact on the drone 2-GCS communication's performance. More also, while encrypting the communication control data, latency difficulties arise. Furthermore, employing secure-based routing protocols raises computational overhead and creates delays. Therefore, there is an urgent need for more research on how to counter drone network and communication protocol issues to prevent malicious attacks, especially during flights. Meanwhile, most of these communication issues are discussed subsequently.

5 Sensor-level issues

Currently, at the sensor level, the variety and complexity of drone onboard sensors, such as physical, chemical, mechanical, and other types, make them targets for enemies. Due to the particular properties of drones, existing countermeasures against sniffing, spoofing, or jamming onboard sensors are inadequate. In the context of drones, existing security research addresses sensor-based risks and attacks [262]. When safeguarding sensed data against malicious actors, additional criteria for drones must be considered, such as energy consumption, the legitimacy of sensor readings, and processing costs.

5.1 Drones use sensors to collect data

Sensors are used by drones to collect data about their surroundings. These are critical data that must be safeguarded against hostile attackers. Compromised drone sensors could lead the drone system to malfunction in hostile situations. Hence, a novel study is urgently needed on how to address the complexity of onboard sensors, such as their physical, chemical, and mechanical mechanisms, which make them vulnerable to cyber-attackers. The following sections detail several sensor-level weaknesses, threats, and possible drone attacks which include;

- (i) *Sensory-channel attack* Drones have a suite of sensors that allow them to attack using their sensory channels, such as infrared, auditory, and light. The study by Son et al. [263] showed that drones with micro-electro-mechanical systems

(MEMS) gyroscopes can fail as a result of deliberate sound disturbance. MEMS gyroscopes resonant at audible frequencies, according to the research. Another study found that changing the immediate environment can damage optical flow camera sensors, which are used to steady drones [67].

(ii) *Security and privacy criteria requirement for drone sensors*

- (a) *Confidentiality* Sensitive data and data interchange between drones and the GCS must be protected from unwanted access, as they could be a source of sensitive flight mission information leaks, such as telemetry data and control commands. One should consider developing robust cryptographic methods to prevent the adversary from gaining such information.
- (b) *Integrity* The necessity of maintaining data integrity cannot be overstated. It's a requirement for a flying mission's success, and it keeps enemies from faking network traffic. A compromise of the system's integrity could cause the drone system's behavior to alter, resulting in mission failure. As a result, any communication must be safeguarded and verified. This criterion can be met using authorized encryption methods [22, 206, 207, 210, 236].
- (c) *Availability* Drones should be able to operate uninterrupted, whether deliberately or inadvertently. All resources required for a flight mission must be accessible to authorized users. In addition, the drone system must be able to survive classic Denial of Service (DoS) attacks, which could jeopardize its availability. IDS can aid in the prevention of such assaults [60].
- (d) *Authenticity* The authentication process is a crucial step in establishing a secure connection between the drone system's many components. It enables the legitimacy and identity of drones engaging in a flight mission to be verified. We use authentication to ensure that each drone is trustworthy, and only authenticated drones can take part in the flight mission. Furthermore, the authentication secures the drone network from spoofing genuine nodes by enemies.
- (e) *Non-Repudiation* Users cannot dispute their behaviors, which include sending and receiving data over drone networks. In the

event of a mission failure, one may have to deal with concerns about accountability. This property prevents the user's operations from being denied. In addition, the drone system must create adequate nonrepudiation methods, such as the digital signature of shared messages.

- (f) *Authorization* Only authorized individuals should have access to the drone system's data. Unauthorized users are not permitted to interact with the drone network in any way. Aside from that, the drone system must define which resources an authorized user has access to. Access control policies must be used to monitor who has access to such resources.
- (g) *Non-disclosure* Indeed, sensitive data shared between the GCS and the drone, such as recorded photographs and video recordings, should not be made public [144]. That is, one must consider the non-disclosure property in the privacy requirements for drone systems in addition to the above-mentioned security criteria.

(iii) *Potential threats to drone systems*

In a drone system, it is possible to have different vulnerabilities for each component of the system. Therefore, the potential threats that may occur for each component may differ. The threats that can occur for each component of the drone system are classified by the security objective [10, 110, 126, 200]. Potential threats that may occur for each component of the drone system are presented in supplementary Table 4.

The bedrock of modern industrial mechanization is Cyber-Physical Systems (CPS). However, collision, jamming, tampering with packets, denial-of-sleep (DoSL), tampering with data, spoofing identity, and repudiation of origin are examples of unauthorized interference in the CPS's functionality at the physical level, which reduces confidentiality in drone flight missions. Network attacks undermine the system's confidentiality, integrity, and availability. Only authorized users have access to information on the system, which is protected by confidentiality. It is possible to eavesdrop on information and fake the system if confidentiality is breached. Integrity refers to the preservation of the original information and signals that have been transferred, stored, and changed. Message injection, replay attacks, and other attacks are possible when integrity is violated. The system's availability permits it to work for the amount of time specified by the user. Service must not be disrupted, and performance must be

sustained in terms of maintenance. In addition, when it comes to system access, the service must be available whenever the user needs it. Availability can be harmed by denial-of-service attacks.

Every drone has its own set of onboard sensors. Even with the strong architecture of drone sensors, the Flight Controller is unable to discern between fraudulent and valid sensor inputs. To construct a secure drone flight mission, major security and privacy criteria are required.

5.2 Existing countermeasures against sensor-based drone attacks

Sensor-based attacks countermeasures: The authors, Wu et al. [301], presented a way to mitigate GPS jamming attacks by permitting autonomous navigation when the flight controller does not receive GPS signals. Other ways focus on detecting anomalous signals using machine learning-based IDS to detect known and unknown sensor-based threats to prevent the adversary from launching a GPS spoofing attack [29, 298]. These systems gather training datasets from the drones' onboard components, such as sensor readings and flight logs.

This study highlights existing countermeasures against sensor-based drone cyber-attacks.

- (i) *Sensor Vulnerabilities and Threats* Drones are highly sensor-dependent gadgets. They are equipped with cameras, a global positioning system (GPS), and accelerometers, among other sensors. Thus, they rely on sensor readings to function effectively. However, because these sensors handle sensitive data, a malevolent operator could use them to jeopardize the flight operation. Civil GPS transmissions, for example, are unencrypted and unauthenticated. As a result, an adversary can take advantage of this flaw by imitating a GPS signal to fool the operator. Exploiting the real-time data from onboard sensors could cause the drone system to malfunction from an attacker's perspective. This attack could occur because the flight controller does not verify the validity of sensor readings. Malicious software can be used to introduce sensor flaws into the drone system as well. This class of vulnerabilities presents a new attack vector enabling the adversary to totally control commercial drones due to the practicality of sensory-channel attacks in real-world situations [262, 284].
- (ii) *Sensor-based exploits* This includes GPS data jamming, bogus sensor data injection, and sensory-channel attacks, among others.

- (iii) *Data jamming via GPS* The onboard GPS receiver gets its GPS location from the satellite and communicates it to the ground control system during a flight mission. A GPS data jamming attack happens when an enemy interrupts the GPS signal's navigation feed, causing the drone to get disoriented [245]. When such attacks are carried out, the drone's control is lost, and the drones may be hijacked.
- (iv) *False sensor data injection* External sensors, such as electro-optical and infrared sensors, can be compromised by injecting fake sensor data readings into the Flight Controller [57]. The drone's stabilization is thwarted as a result of this compromise. By gaining access to the onboard Flight Controller system or manipulating sensor readings through system calls, an attacker can inject fake sensor data into drones. Otherwise, attackers can immediately send fake signals to the sensors, putting the flying drone at risk. GPS spoofing is a popular example of false sensor data injection attacks. Since GPS signal broadcasts are frequently unauthenticated and unencrypted, the attacker seizes the vulnerability to deploy a spoofing attack on the GPS to modify the drone's GPS receiver by mimicking the generated signal [293]. Therefore, the intruder obtains control of the drone. The studies by Seo et al. [248] and Kerns et al. [136] demonstrate a GPS spoofing exploit on drones. The drone can be forced to respond to fake signals as a result of the GPS spoofing attack, which disrupts its navigation system. Meanwhile, security issues at the sensor level, existing solutions, countermeasures, and their limitations are presented in supplementary Table 5.

However, due to the drones' limited energy and computational resources, real-world application is difficult. When GPS signals are unavailable, Brown and Lu [45] recommend the use of extra sensors as an alternate navigation method. To enable autonomous drone flight under a loss-of-GPS scenario, the scientists paired a monocular camera visual sensor with an inertial measurement unit (IMU) sensor. To avoid introducing fake Flight Controller sensor data, obtain readings from a different set of sensors and cross-verify the data. Another way to detect external sensor attacks is to use a control-invariant technique to represent the drone's physical attributes [59]. The control invariant technique verifies that the drone's physical state is consistent with its predicted state, as determined by its control model. In a similar vein, Quinonez et al. [229] presented an architecture for securing sensor readings in

the presence of physical invariants. Meanwhile, most of the hardware issues are discussed as follows.

6 Hardware issues

Right at the hardware level, drone hardware could be targeted even in the manufacturing process, before or during the flight mission, regardless of the kind and characteristics of distinct commercial drones, such as hardware and firmware type. These scenarios are feasible because of flaws in drone firmware as well as the lack of encryption in bespoke chipsets. Given the widespread use and variety of existing drones, it is critical to develop a consistent hardware security strategy for defending drones against hardware-based cyber-attacks. Therefore, there is an urgent need for more research on how to counter hardware attacks, especially during flights. Meanwhile, most of these hardware issues are discussed as highlighted in sections (a) to (c).

6.1 Hardware-level issues

Drones are seen by opponents as a possible tool for carrying out physical strikes in national airspace. The onboard flight controller (FC) and the ground control station (GCS) are the hardware components of a drone system. Both hardware devices have security vulnerabilities that could lead to cyber or physical attacks. This section is dedicated to presenting the vulnerabilities that an attacker can use to compromise a drone system's hardware. Then, to reduce hardware-based threats, we present the available protection measures as follows:

- (i) *Hardware Vulnerabilities and Threats* Hardware Trojans, physical drone collisions, flying skill concerns, and hardware failures are all examples of hardware-level vulnerabilities and attacks.
- (ii) *Hardware Trojans* Modifications to electrical hardware, such as enlarging the logic gate and messing with the hardware circuit [182], are common in hardware trojans. Hardware trojans specifically target the flight controller, making the drone system open to a variety of attacks. A non-trusted third party maliciously embeds the hardware's software, such as trojans in the Flight Controller's semiconductor supply chain [230]. The attacker uses these alterations to compromise the FC's integrated circuit and security features, such as by leaking the flight controller's cryptographic keys, slowing down the propeller rotation speed, and so on. The Actel ProASIC chip in the Boeing 787 jet was revealed to have a trojan [52].

The backdoor allows the attacker to monitor and operate the avionics system, risking the safety of those on board.

- (iii) *Physical drone collisions* Physical collisions between drones could occur during a flight that requires collaboration and cooperation between numerous drones, resulting in the drones crashing. Drones rely largely on collision avoidance systems (CAS) to avoid such collisions in public airspace [20, 119]. These systems, on the other hand, lack built-in security safeguards and are unable to address the problem of malevolent actors causing collisions [11, 108].
- (iv) *Hardware failures* Drones can have issues with their hardware components, such as motor problems or battery life. These technological problems pose a risk to the flight mission and may result in the drones landing in an unsafe area [25]. If the drones store unencrypted data in this situation, the adversary can reveal crucial mission-related data and compromise the flight mission's anonymity.
- (v) *Flying skills issues* These challenges arise when semi-autonomous or non-autonomous drones are controlled remotely by humans, particularly those that are very sensitive to wind disturbance due to their complicated dynamics and size [157]. They necessitate flying abilities such as remote control of the drone's speed, height, and direction. In such cases, the operator's lack of technical capabilities could cause the drone to crash, resulting in an operational failure. Therefore, the drones are vulnerable to physical theft.

7 Hardware-based attacks

Hardware-based attacks include radio frequency modules, hijacking, battery attacks, and supply chain attacks, which is further discussed as follows:

- (i) *Hijacking* Drones, by their very nature, are visible at low altitudes, making them ideal candidates for hijacking. The attacker uses malicious software to take control of a flying drone, either remotely or directly. The use of anti-drone guns is a simple method for disabling and hijacking drones [118]. They are frequently in law enforcement's custody to keep malicious drones from loitering in restricted flying zones. Nonetheless, the attacker can ground or hijack the drones with the same firearm.
- (ii) *Supply chain attacks* With the expansion of the drone industry, attackers now have a larger

opportunity for compromising drones through supply chain cyber-attacks. This sort of attack targets less-secure and sensitive components such as airframes, propellers, and actuators by exploiting vulnerabilities in an organization's supply chain process. Thus, the final product given to the consumer is already tainted. The research by Belikovetsky et al. [36] demonstrates a realistic supply chain attack against drones. The researchers used additive manufacturing (AM) to carry out a physical supply chain attack on drones. The attack entails remotely changing the propeller design files to sabotage a specific drone. During a flight operation, the enemy diminishes the fatigue life of the 3D-printed propeller and causes delayed damage. This Belikovetsky et al. [36] research demonstrates that detecting sabotage attacks in additive manufacturing systems is still a difficult research area that requires in-depth exploration.

- (iii) *Battery attacks* Lithium-ion rechargeable batteries are commonly used in drones. The battery management system (BMS) helps these batteries provide dependable energy to the drone system's components. However, an opponent might deplete the battery's energy by launching hypothetical battery depletion attacks [71]. As a result, the drone system malfunctions, jeopardizing the batteries' availability, secrecy, and integrity [171]. By physically tampering with or swapping genuine batteries with bad ones, the attacker compromises the availability of drone batteries, causing the drone system to fail. Another attack option is for the enemy to cause a deep discharge of the batteries. This type of attack might be carried out by compromising other drone components, such as by injecting malicious software or spoofing sensors, resulting in the drone batteries being depleted [165]. Modifying real battery information to the operator via the drone-2-GCS data exchange is one way to compromise the integrity of drone batteries. Furthermore, disclosing sensitive battery-related data such as the state-of-charge (SoC), which shows the ratio of available charge to drone battery capacity, can jeopardize the confidentiality of drone batteries.
- (iv) *Radio frequency module attacks* Radio frequency (RF) modules are used to send and receive radio signals between two devices. In the case of drones, an operator could send control signals to the flying drones using a standard remote controller or the GCS. The enemy can jam the control signals and deactivate the drone-2GCS communication in this situation, leaving the drones in a lost-link status.

Mekdad et al. [182] investigated the XBee 868LP protocol, a low-power radio frequency module used for drone-2-GCS communications and showed a replay attack on it. Using a third XBee chip, the attacker modifies the drone-2-GCS communication in this attack. By combining existing features of the chip to access the address of the XBee communication channel, the attacker undermines the security of the primary XBee's communication channel.

7.1 Countermeasures for hardware-based attacks

Physical protection measures should be examined and improved to counter the physical vulnerabilities and hazards posed by drones. Developing ML-based intrusion detection systems to detect such hardware attacks is one viable mitigation strategy for ensuring a trojan-free drone. Using IDS solutions, you can detect the existence of tampered data or commands by: (a) learning a model based on the average data supplied by the pulse width modulation (PWM) signals. These signals are typically found in drone ICs. (b) Using malicious data to train the model. These data are obtained by exploiting firmware flaws or inserting trojans into electronic devices. As a result, the PWM signals are affected. A fine-grained circuit analysis is another mitigation strategy that can be used to detect hardware trojans [197].

Using authorized encryption to protect both the GCS and the drones from unauthorized access and keeping them malware-free will dramatically reduce the chances of hostile actors taking control and hijacking the flying drone. Furthermore, modifying the flight paths could hinder the enemy from recognizing the flying pattern, making physical theft of the target more difficult. McNeely et al. [179] proposed a drone hijacking detection system based on a statistical analysis of conventional flying patterns in a study. The efficiency of their detection algorithm is demonstrated through simulations of various hijacking scenarios. Nevertheless, when simulation factors such as control instability are modified, their method fails, prompting more testing and enhancement of the simulation data quality.

In order to avoid employing hacked drone components, supply chain attacks can be minimized by monitoring supply chain security during the production process [299]. Furthermore, tamper-proofing technologies such as tamper-proof microprocessors, anti-tamper software, and so on. It will prevent illegal physical or logical modifications that could compromise the drone's vital components' validity. The inclusion of safety circuits in the battery management

system, which offers physical battery protection for drones, is one existing countermeasure to prevent battery depletion attacks. Furthermore, a pre-flight battery evaluation would be a fair method for ensuring a safe flying mission. Another option, which involves monitoring the battery discharge process in real time, could detect depletion attacks during the flight operation. Notwithstanding, if the UAV-2-GCS data transmission is not verified, the attacker may forge the transmission and show the operator an inaccurate battery level. As a result, cryptographic techniques must be used to secure the UAV-2-GCS data transmission. Furthermore, we can utilize machine learning techniques to detect known drone battery depletion threats. The features extracted from simulated battery depletion attacks [71] can be used to accomplish this.

The manufacturer can use the chip's inbuilt encryption to protect against radio frequency module attacks. Yet, this technique is limited because it reduces the chip's bandwidth and increases its latency. In this situation, the authors [182] advised that the encryption be outsourced to a different chip. Although this solution ensures the confidentiality of data transferred via the radio channel because remote commands are sent directly to the chip, however, it does not prevent the attacker from executing them. Hence, by assigning random values to destination addresses, the attacker can launch a DoS attack. Encrypting the radio control (RC) link is another option. In 2017, Podhradsky et al. created an encrypted RC connection that is compatible with resource-constrained devices using the Galois embedded crypto (GEC) library [292]. The proposed solution allows the drone and the RC transmitter to communicate in a secure manner. Supply chain attacks, hijacking tactics, radio frequency modules, and battery depletion attacks are just a few of the existing attacks against drones on the hardware level; their countermeasures and limitations are shown in supplementary Table 6.

Many research communities have proposed security solutions to drone hardware-level issues, which include building defense mechanisms at the hardware level. Security solutions are provided for hardware-covered areas such as the battery management system, by applying safety circuits, doing a fine-grained circuit analysis, and regulating the supply chain's security during the manufacturing process. Despite the fact that current countermeasures try to safeguard drones from hardware-based attacks, there are still some constraints to consider. For example, hardware encryption or obfuscation techniques can obstruct fine-grained circuit examination; built-in encryption on RF modules reduces bandwidth and increases chip latency. Furthermore, security implementations are not considered in the creation of collision avoidance systems (CAS). Therefore, there is an urgent need for more research on how to provide some countermeasures for drone hardware-

based attacks, especially during flights. These countermeasures could include new encryption techniques for securing hardware like RF modules, battery depletion, supply chains, hardware hijacking, hardware failures, physical collisions, and so on.

8 Software issues

After the discussion on hardware-level difficulties, the software issues and the software-level-based issues can also be discussed, such as the aspect of vulnerabilities, threats, and attacks that target drone software. Software issues include malfunctioning, hacking, and operating system problems, which is further discussed as follows:

- (i) *Software malfunction* Many drones have previously launched weapons toward civilians, resulting in a considerable number of casualties, injuries, and property damage owing to faults or software mistakes. Drone mistakes also endanger the lives of other military personnel. Drones are continually being improved in order to reduce mishaps or hazards that could jeopardize human health and safety.
- (ii) *Easy to hack drones* One significant disadvantage of drone technology is its susceptibility. Hackers can swiftly get access to a drone's central control system and assume control of the drone. Without the initial operator's understanding, the principal control system has important knowledge that hackers can exploit. Hackers can get access to personal information, corrupt or damage files, and expose information to unauthorized third parties.
- (iii) *Operating system attacks* The system software of the flight controller could be used to launch attacks against civilian or military drone missions. The drones and their payloads will be lost if the system software is corrupted.

Amazon's Prime Air service, which uses drones to deliver packages, is an example of a civilian application that could be vulnerable to operating system attacks [12]. Attacking the delivery system could cause the recipient's delivery cargo to be delayed, affecting the business's performance and credibility. Attacking drone operating systems entails remotely inserting malicious software into drones, such as Maldrone, and then, seizing control of the system to hijack the drone. To accomplish this, the attacker can obtain the FC's cryptographic key and steal the unencrypted data stored on the device.

- (iv) *Reuse of drone apps* Android apps are compatible with the drone's apps. It is feasible that these apps are associated with drones that use the same underlying firmware as the CVE-2017–3209-vulnerable DBPOWER and USA Toyz devices. Anyone may control and receive video streams from USA Toyz drones using the DBPOWER/RC app for DBPOWER drones. DBPOWER drones can also be controlled via the USA Toyz Flying app. The explanation for this could be that the same vendor, Udir Toys Industrial Company Limited, is responsible for both the DBPOWER/RC and Flying apps. However, other apps from different or the same manufacturers in the Google Play Store can be used to fly the Discovery drones and view their videos.

9 Software-based issues

Software-level issues, such as the aspect of vulnerabilities, threats, and attacks that focus on or target drone software, are succinctly discussed, followed by how to counter such attacks using existing defense paradigms.

At the software level, terrorists can undermine the flight mission by exploiting zero-day and current software vulnerabilities in the flying stack and the GCS software. The ubiquity of software-based threats highlights the necessity for drone software security protection solutions to be developed. Existing drone manufacturers, on the other hand, resist incorporating software security mechanisms into their devices due to performance concerns. As a result, adversaries can use this flaw to create malicious software like SkyJack [64], Snoopy [72, 215], and Maldrone [214]. Therefore, there is an urgent need for more research on how to counter software attacks, especially during flights. Meanwhile, most of these software issues are discussed as follows:

- (i) *Software Vulnerabilities and Threats* Malicious software and zero-day vulnerabilities are examples of software-level vulnerabilities and cyber-attacks on drones.
- (ii) *Malicious software* Malicious software can infect the ground control station and the flight controller. Drone malware can lead to the loss of sensitive data as well as the loss of control of the drone system. An attacker with access to the drone's flight stack might theoretically shut down the drone system, resulting in a denial-of-service assault and disrupting the flight mission.

Drones' security and privacy could be jeopardized if malware is embedded in them Maldrone

[72], for example, is a virus that infects the flight controller and allows the attacker to control the drone. It acts as a proxy for the drone's flight controller and sensor communications, allowing it to land at any desired place. The hijacking malware SkyJack can be installed on a targeted drone [64]. It can use the Wi-Fi de-authentication attack to take control of other authorized drones and undermine the entire system. Snoopy is a type of malware that can be installed on a drone and used to steal personal data from other people. It employs impersonation tactics to persuade users to connect to a phony Wi-Fi network.

After that, Snoopy follows its users and collects their personal data. Recently, a new sort of malware has emerged that encrypts or locks a user's data or machine and keeps it encrypted or locked until the user pays a ransom to the opponent. Ransomware is the name for this sort of infection [213]. Though ransomware attacks on drones are still uncommon, one must consider future ransomware targets on drones in order to provide countermeasures.

- (iii) *Zero-day vulnerabilities* Unexpected vulnerabilities, such as DoS or buffer overflow attacks, may exist in the drone's flying stack or GCS software. The manufacturers of the drones are unaware of these flaws, which can pose a serious threat to the operators. Until the drone's manufacturers issue suitable updates, enemies can continue to exploit zero-day vulnerabilities. Operators, on the other hand, must upgrade their drone devices for each patch released.
- (iv) *Software-based Attacks* Operating system attacks, manipulating captured films, and system ID spoofing are examples of software-based drone attacks.
- (v) *Attempting to tamper with captured videos* During a flight operation, the operating system employs system calls to capture movies from the onboard camera in order to ensure safe navigation and avoid crashes [68]. A knowledgeable attacker with access to the system settings, on the other hand, can intercept and hijack the sent system calls. To control the flying drone, the adversary could combine the tampering attack with a GPS spoofing attack. The adversary's primary purpose, unlike operating system attacks, is to jeopardize the navigation's safety and cause crashes.
- (vi) *System ID spoofing* Drones communicate their System ID and location to third parties such as government agencies and law enforcement when required, according to FAA regulations [48, 278].

But, because most existing drones lack encryption, an attacker can pose as a third party and use an identity spoofing attack to breach the communication channel and obtain the drone's system ID [60].

9.1 Countermeasures for software-based attacks

Drones and their payloads can be kept safe by updating their operating systems on a regular basis. Furthermore, firewall systems on the GCS can prevent malicious communication from reaching the drones. Additionally, software-based solutions such as antivirus and intrusion detection systems (IDSs) can monitor network traffic and protect drones from harmful activity. However, due to computational and energy constraints, implementing an onboard IDS is difficult. Additionally, implementing permission methods for drone system resources can aid in the prevention of malicious code execution. The adoption of software-based authentication methodologies is a viable solution against software-based assaults. They verify that the software running on the flight stack is safe [79, 250]. Remote authentication solutions are low-cost and ensure that the software stack is legitimate.

The adversary uses malicious software and zero-day exploits to infiltrate the flight stack at the software level. Furthermore, the opponent can tamper with the acquired videos in order to deceive the operator. Antivirus and intrusion detection systems (IDS) are used to protect against software-based threats. In addition, the operator should keep his operating system up to date and use software-based attestation techniques to ensure that the code running on the operating system is legitimate. However, the proposed security methods against software-based threats are insufficient to effectively protect the flight stack from malicious attacks, and for publicized zero-day vulnerabilities, the patching procedure can take many weeks. Therefore, the drones are vulnerable to attackers. Additionally, deploying IDS or firewall implementations on the GCS can raise computation costs and cause latency concerns. Drone software security concerns, current countermeasures, and limitations are summarized in supplementary Table 7.

9.2 Security-performance tradeoff

When deploying security solutions at any level of the drone, it is necessary to examine the system's performance, that is, the calculation costs, communication expenses, energy costs, and storage overheads in particular. Adding an extra security layer for each level without taking into account the aforementioned parameters, on the other hand,

could reduce the aircraft's performance. At this stage, one can deduce probable tradeoffs between security requirements and drone performance.

10 Physical attack issues

Physical attack issues with drones include drone destruction, payload destruction, civilian or military person loss of life, property or infrastructure destruction, and instigation. Likewise, physical attacks can be the result of remotely accessing someone else's drone, which has a vulnerability to wild animals, is weather-dependent, and can deploy added payloads such as nuclear, biological, radiological, and chemical primitives. These attacks can be further discussed as follows:

- i. *Destruction of Drones* Drones, unlike commercial planes and helicopters, cannot transport large payloads or deliver commodities over long distances. Hence, it will be insecure if drones are used to carry heavy object(s) that can cause damage.
- ii. *Destroy payload* The payload is the maximum amount of weight that a drone or unmanned aerial vehicle (UAV) can carry. It is usually calculated separately from the drone's weight and includes anything added to the drone, such as extra sensors, cameras, or delivery goods. If a payload is too much for the drone, it can damage the drone. There is a need to secure the drone and the payload during flight so that both will not be destroyed. More also, there is a need to determine the weight of the drone. Then, to ensure that your drone will be able to hover, add 20% to the total. For example, if a drone weighs 1 500 grams, then one needs 3 000 grams of force to lift it off the ground, plus another 600 grams to hover it. The thrust total is the number that is needed. Furthermore, divide the total thrust by the number of motors required by the drone. Therefore, if the total thrust is 3 grams and the drone is a quadcopter, each motor will require 900 grams of push.
- iii. *Drone swarms will almost certainly be very effective for mass killing, whether civilian or military* They could be used as assassination weapons by terrorists and as strategic deterrent weapons by some countries without nuclear weapons.
- iv. *Destroy property or infrastructure* Although drones can use their remote control capabilities to monitor sites, communicate potential hazards,

and alert people to potentially dangerous situations, they can also be used for attacks on critical infrastructure such as oil and gas, refineries, pipelines, flare stacks, and energy transmission lines.

- v. *Instigation* Drones can capture data, information, images and human pictures while remaining undetectable.
- vi. Deploys added payload (nuclear, biological, radiological, chemical, etc.).
- vii. The issues of remotely accessing someone else's drone to take images or video of a private setting, stealing a drone mid-flight, and taking down a drone operated by someone else [288] need to be resolved.
- viii. *Vulnerable to wild animal attacks* Drones are vulnerable to wild animal attacks and can also be damaging to the environment. When a drone operator is flying in an area with a large number of wild animals, it is likely that they will crash into a tree or collide with a vulnerable animal.
- ix. *Weather dependent* When compared to regular aircraft, drones are more subject to weather conditions. For example, if the weather is bad, the drone will not be able to maneuver properly or collect reliable data or images. Likewise, most drones cannot resist windy or stormy conditions, including heavy rain. However, there are drones that are more stable and can successfully survive wind gusts.
- x. *Fly-away attacks* include catering or escaping with the drone.
- xi. *Shooting down a flying drone* creating accidents such as the one recorded in Seattle.
- xii. *Lock-out attack* prevents the device's lawful owner from connecting to it.

11 Emerging security issues

Nowadays, there are other emerging areas of drone security threat, such as terrorist attacks, mid-air collisions, illegal surveillance, smuggling, electronic snooping, and reconnaissance. These areas are discussed as follows:

- (i) *Drone terrorist attacks* Drones are attractive to terrorists for a number of reasons. The first reason is that drones have intrinsic mobility. Drones enable attacks to be carried out over perimeter fortifications [81]. Terrorists could use drones to carry out attacks easily and with great success. That is, a drone can be used (i) to attack targets that are difficult to reach

by land (ii) to carry out a large-scale attack aimed at inflicting a maximum death rate on a population (iii) to carry out a wide-scale attack aimed at inflicting a maximum death rate on a population (iv) The accomplishment of long range and acceptable accuracy with relatively inexpensive and increasingly available technology (v) Ineffectiveness of existing air defenses against low-flying drones (vi) Drones' relative cost efficacy when compared to ballistic missiles and manned planes (vii) Possibility of inducing a big psychological effect by frightening people and putting pressure on politicians. A terrorist might equip a drone with guns or explosives and fly it onto people or structures to cause physical harm or death. Individuals, buildings, or transportation infrastructure, such as commercial airliners, may be targets of these attacks [31, 50, 62, 187].

- (ii) *Drone mid-air collisions* Drones air-colliding issues with manned aircraft can inflict more structural destruction than birds of the same weight, at a given impact speed. Most drones are built of more rigid materials than birds, which have soft and flexible tissue. The stiffest components of the drone, such as the battery, motor, and payload, were shown to inflict the most damage during testing. The researchers discovered that concentrating those masses on the drone could do more damage. Naturally, birds possess self-preservation mechanisms that allow them to escape mid-air collisions in most circumstances, and this is exactly what the detect-and-avoid systems are designed to do. A major concern for further research is the ability to use a variety of technologies that would allow drones to get out of the way of other flying aircraft. Although a mid-air collision is uncommon, there are some characteristics that can be seen in the majority of mid-air crashes. Contrary to popular belief, mid-air collisions rarely occur in foggy weather or at night when visibility is reduced. Perception is the mechanism of processing the data provided by the sensors to identify hazards, whereas sensing is the actual sensory equipment, such as a camera. The systems can be passive, that is, no energy is emitted; active, which means energy is bounced off a target, or a blend of both. Therefore, urgent research into sensors used in collision avoidance systems is

required. More importantly, there is a need for effective algorithms that can be used in sensors to derive criteria for systems suitable for implementation on drones to avoid mid-air collision.

- (iii) *Illegal surveillance* Drone surveillance from the sky poses a threat to human privacy. Drones may be outfitted with a variety of surveillance systems that capture high-definition video and still images at all hours of the day and night. Drones can be outfitted with equipment that allows them to listen in on phone calls, track geographic positions, and collect license plate data. Some drones, such as the RF drone, use radio frequency (RF) sensors that automatically listen and monitor the 70 MHz to 6 GHz bands for transmissions of the communication link, that is, between the drone and the receiver, to track the location of the drone. Drones are now able to monitor large metropolitan populations in real time. Law enforcement now uses breakthroughs in video technology and biometric identification to track individuals across cities and in crowds. Aerial surveillance for business is a developing sector, but it also poses a growing privacy danger. In the USA, the Federal Aviation Administration was ordered by Congress in 2012 to incorporate drones into the National Airspace, but there are still no clear standards safeguarding individual privacy from drone surveillance. Drone regulations should include (i) drone ID requirements, such as broadcasting the drone's name and location, as well as information about the drone's purpose and technological capabilities, and, if applicable, the government or commercial operator. (ii) A prohibition on the government or government contractors conducting widespread aerial surveillance. (iii) The use of drones by the government is subject to a warrant requirement. (iv) Restrictions on data collection for individual and commercial purposes. (v) Transparency criteria for individual, commercial and government operators.
- (iv) *Drone use for smuggling* Nowadays, drones are being used to commit different crimes, which include smuggling of drugs, weapons, foods, etc. For instance, in the USA, inmates at the Lee Correctional Institution do not need to know about Ellis Redding. Instead of the fictional prison contraband courier in Stephen

King's novel, narcotics, candy, and weapons can be supplied by drone to the Bishopville, South Carolina facility. Drone-borne contraband smuggling has become a rising problem in jails across the USA and worldwide. The US Bureau of Prisons publicly sought technologies to detect and intercept them in 2015, and the UK government developed a dedicated squadron to "address the threat of drones to prisons" in 2017. According to the state correctional department's director, the problem has been particularly bad in South Carolina, where incursions occur every week. Narcotics, cash, candy, phones, firearms, and weapons have all been delivered late at night at Lee Correctional Institution. Violence is a serious possibility when it comes to illegal drugs. Seven convicts were killed in a riot at the Lee jail in 2018. In another incident at the site in 2010, a gang member used a smuggled cell phone to order a hit on the Contraband Unit's head, who was ultimately shot six times at his residence. However, most recently, drone smugglers have been charged by the US Justice Department. Nearly two dozen suspects were arrested and charged with a variety of offenses, including contraband, criminal conspiracy, and illegal narcotics possession. So far, 12 drones have been confiscated, including two during the most recent arrest. According to Sheriff Daniel Simon, they were among the largest ever seen, indicating that they could transport heavier cargo.

- (v) *Electronic snooping* A flying drone can get into Wi-Fi networks and spy on people's phones. In some cases, federal laws have declared that electronic snooping is a kind of fraud involving the use of computers. Information is stolen, and terrorism can happen as a result of cyber-attacks and computer-related criminality, thus, individuals who snoop in this manner may face felony penalties. A snooping attack is when an attacker listens in on traffic between two workstations on one's network. An unauthorized person could potentially enter one's network and read confidential data if traffic includes transferring unencrypted passwords. The distinction between eavesdropping and snooping is that to listen in while snooping is to be crafty and cunning so as not to be noticed; to listen in while snooping is to be tricky and cunning so

as not to be seen. On the contrary, a drone can be utilized to perform great things, such as taking measurements. For instance, a drone can be equipped with almost any type of sensor. A power plant in Japan, for example, was damaged by a tsunami (a series of powerful ocean waves) in 2011 and began spewing radioactive material. People should not enter the area since it is still unsafe. Scientists, on the other hand, have sent in drones to assess radiation levels in the area. In 2013, NASA scientists outfitted three drones with electronic sensors to sniff for sulfur dioxide as they flew over the rim and through the plume of a Costa Rican active volcano. Interception, jamming, and packet injection are three areas of electronic attack where commercially available software-defined radio (SDR) platforms, such as the Hack RF, can constitute a threat.

- (vi) *Drone use for reconnaissance* Nowadays, drones are being used secretly to gather information about a targeted organization or government. Reconnaissance exploits are a type of attack that gathers information about a target. That is, reconnaissance attacks are used to obtain information on a specific network or system. These kinds of attacks can take both intellectual and physical forms. Reconnaissance attacks are in phases, which include: open source intelligence (OSINT) gathering, foot-printing, human recon, verification, and the vitality phase, which is to confirm the reachability of the IP addresses identified in prior phases. This is a transitional phase between reconnaissance and enumeration. Reconnaissance attacks are dangerous, and they should be treated as a serious threat to an organization since they may provide potential attackers with the information they need to launch access or DoS attacks. The US Marine Corps Force Reconnaissance created two key mission functions during the Vietnam War in order to grow and perfect this capability. They began by concentrating on issues that had less to do with argument and confrontation with hostile opponents. These types of reconnaissance missions are referred to as “keyhole” or “green” ops by the US Marines. These operations, as well as the accompanying tactics and techniques used throughout the operation and mission, were designed to perform deep reconnaissance tactics. The job

entailed simply identifying, gathering, and collecting all relevant military intelligence while also observing, identifying, and reporting adversaries and key data about them. However, these reconnaissance attacks can be prevented, whether the information is collected through network probing, social engineering, or physical observation.

- (vii) *Use trained eagle birds to capture drones* Drones operating in their space are routinely attacked and even captured by large flying birds, such as eagles. At a French Air Force facility in Southwestern France, these drone-hunting birds of prey are being trained. They are literally born on top of drones, and they are kept there until they are ready to feed. They are taken to a field to intercept drones once they are ready to fly. Consequently, the eagles are rewarded with meat as a result. An illustration of how the French Air Force trains eagles to hunt drones is shown in supplementary Fig. 3.
- (viii) *Cyber espionage* Cyber espionage is a significant threat to drone development; various cyber espionage campaigns have been conducted in recent months to steal technological intellectual property from defense manufacturers. UAVs are considered primary targets in cyber-espionage; the technique used is consolidated; attackers conduct spear-phishing attacks with the goal of stealing technology-related information, such as UAVs. FireEye, a security firm, revealed in early 2013 that it had discovered an APT campaign targeting companies in the defense and aerospace sectors that originated in some country and was designed to steal intellectual property and industrial secrets from US companies. “The set of objectives covers all aspects of unmanned vehicles, including land, air, and sea vehicles, from research to design and manufacturing of the vehicles and their various components.” Dealing with the phenomenon of cyber espionage requires a structured approach; the threat has grown in complexity in recent years, and attackers have been able to evade the main security mechanisms for extended periods of time in many cases.

12 Privacy issues with drones

The advancement of drone technology has put a wide range of privacy concerns at risk. The privacy issues in this area are classified according to the entity at risk and the sort of sensitive data that is revealed to unauthorized users. Hence, this section presents the categorization of privacy issues with drones, first highlighted in supplementary Fig. 4.

These main privacy issues of drones highlighted in supplementary Figure 4 are discussed as follows:

12.1 Drone privacy issues

Drones can violate privacy rights, from spying on people's lifestyles to acquiring sensitive data about an individual or organization, in addition to raising security concerns. Individuals' privacy is jeopardized by the operation of drones in civilian airspace without regulations. Furthermore, sensitive data captured by drones and relayed to the GCS must be kept safe from prying eyes. As a result, privacy leaks must be considered while designing drone systems. Attacks on privacy are possible due to two fundamental scientific gaps:

- (i) *The detection and identification problems* The detection challenge entails determining the difference between a legitimate nearby drone and a malicious drone that invades an individual's privacy. Existing methods to address the detection problem are limited since they are unable to identify spying on a single point of interest (PoI) [42, 53, 239, 271]. A recent study demonstrated using a cryptanalysis approach, that by applying a periodic physical stimulus (LED flicker) on the spying drone cameras, it causes a watermark on the encrypted UAV2-GCS communication traffic [192]. The identification of such a watermark establishes whether the drone is serving a legitimate or illegitimate function. However, in the drone-2-GCS communication channel, this technique is limited to WiFi First-Person-View (FPV) transmission. Given a multi-drone environment, it is likely hard to distinguish an adversary drone from lawful ones in the identification challenge.

Although methods for determining if a person is a friend or enemy exist [192, 193], However, they are unable to distinguish a nearby enemy drone from a legitimate one with the same GPS location, such as less than 4.9 m in altitude [193, 216]. Thus, malicious attackers take advantage of scientific

loopholes and vulnerabilities to infringe on people's privacy. Therefore, there is an urgent need for more research on how to provide some countermeasures for drone software-based attacks, especially during flights. These countermeasures can include some new encryption techniques to secure operating systems, images, videos, system ID spoofing, and zero-day vulnerabilities.

- (ii) *Privacy Risks* There are three types of privacy risks associated with drones: risks to drones, risks to organizations, and risks to individuals. Sensitive data leaked to third parties are a risk for the drone's category. An organization's privacy risks are connected to organizational data that drones can capture. Individual privacy risks are linked to personal information received by a flying drone. Data privacy is when the secrecy of data that should not be communicated with third parties is compromised.
- (iii) *Risks for Individuals* The usage of drones in aerial photography has raised privacy concerns, according to recent research [129]. In a study published in 2020, Yaacoub et al. categorized privacy leaks into three categories: behavior, physical privacy, and location privacy.
- (iv) In physical privacy, the attacker invades people's physical privacy by capturing images, photographs, and videos of people inside their homes for malicious intentions [185]. Since the major physical privacy concern is the use of drones to spy on individuals, therefore, regulations limiting the use of drones in civilian airspace are required. Drones with directional microphones can also be used to listen in on private talks.
- (v) *Location privacy* People's physical whereabouts and movements are targeted without their knowing that they are being watched [86]. It could be used by third parties for commercial purposes, for instance, to target advertising by location. Drones are now being used for more than just aerial surveillance, they are also being used to track humans anywhere [178]. However, it is difficult to determine whether a flying drone is being used for aerial surveillance or for tracking people [178]. Currently, detecting such invasions of privacy is still a work in progress.
- (vi) *Behavior privacy* In public spaces, the attacker observes people's behaviors, and

interests [63]. Individual surveillance through systematic monitoring of their activities or patterns poses a significant threat to behavioral privacy and may have a negative influence on people's psychological well-being [63].

- (vii) *Risks for Organizations* An adversary's maliciously harvested information and resources from drones might or might not be personal [167]. Drones, for instance, may spy on enterprises via video streaming, which is known as industrial espionage, and the adversary can reveal government and corporate secrets to unauthorized entities. An attacker can spy on a farming business that utilizes a swarm of drones to optimize operations and increase crop production. For example, by utilizing the same drone model. One may not be able to tell which drone is benign and which is malicious in this instance. This inability to tell which drone is benign and which is malicious is called the "identity problem," and it has been used to commit crimes like smuggling and terrorism [192].
- (viii) *Risks for Drones* This category includes unauthorized third parties receiving sensitive information monitored by drones, such as video footage, images, and physical measurements. Other sorts of sensitive data associated with flying drones, including real-time GPS location, height, speed, and battery status, must also be kept confidential and accessible only to the operator. Lin et al. [162] opined that maintaining the data privacy of flying drones is a critical need for the mission's safety. In unencrypted communications, the enemy or attacker has the upper hand.
- (ix) *Traffic Analysis Attack* by monitoring traffic and obtaining vital flight plan data, the drone's anonymity and privacy are jeopardized by this form of passive attack. Digital data can be recovered using forensics techniques such as data extraction and analysis, even in encrypted conversations [241]. Another sort of drone privacy assault happens when an opponent obtains unauthorized access to the drone system's vital components, such as sensors and storage, for example, by injecting hardware trojans, and hijacking attacks. The opponent in this scenario leaks flight data to the public, jeopardizing the flight plan.

12.2 Defense mechanisms against privacy risks

Several studies have suggested privacy-preserving mechanisms to prevent leaking secret information to unauthorized parties. These mechanisms include encryption techniques and the design of tamper-proof hardware so that even in scenarios where the drones are hijacked, they cannot reveal sensitive information. In the study by Birnbach et al. [42], the authors suggested an approach to detect privacy invasion attacks based on a drone's flight behavior. However, it fails to identify a drone's purpose (whether it is legitimate or malicious). The research by Chen et al. [58] presented a privacy-preserving authentication scheme for drone control systems. The proposed architecture has a mutual authentication mechanism to secure communication between entities and integrates cryptography mechanisms such as elliptic curve cryptography (ECC), digital signatures, and hash functions. Moreover, the suggested privacy protection protocol guarantees location privacy and proves its applicability in sensitive control areas.

Similarly, a privacy-preserving authentication approach for drones was proposed by Tian et al. in 2019. It is a predictive authentication framework that considers identity, location, and flying routes as sensitive information. Other solutions can overcome privacy issues, such as implementing access policies and lightweight cryptography approaches. Some manufacturers include a list of no-fly GPS coordinates covering sensitive areas in the firmware of their products. Moreover, regular users can register their home location in the NoFlyZone Database (Altawy and Youssef 2016). Therefore, there is an urgent need for more research on how to provide some countermeasures for drone privacy-based issues, especially during flights. These countermeasures could include new rules and regulations regarding risks to people and organizations, behavior privacy, locational privacy, privacy defense mechanisms, traffic analysis attacks, security, and privacy laws. Therefore, there is an urgent need for more research on how to address privacy issues of drones, especially during flights. Likewise, the regulatory issues surrounding drones need urgent attention in order to mitigate proliferation, threats, and the risk of attacks.

13 Drone regulation issues

As drones become more common in the sky, especially in developed countries, privacy concerns arise. Residents in cities around the USA are taking preemptive measures to protect privacy as the number of drones with cameras increases around offices and residential areas. Police in Westport, Connecticut, planned to deploy specialized

drones to monitor if people were socially distancing in the wake of the COVID-19 health crisis [14]. Many states are increasingly taking a look at drone regulations. For instance, as of September 2021, there were 1.7 million drones, 500,000 of them commercial. However, when most flights can take place at a height of 40 feet above ground level in the air, then it will mitigate privacy concerns in society. Currently, some communities in the USA have established a designated landing zone instead of allowing drones to fly from door to door. Some drones can be controlled; for example, Aquiline drones have a mobile app that will let its clients book a drone service on demand, according to Tech News World. That is, they can accurately execute missions, live stream data and videos and get real-time data insights through the computer or phone app. Therefore, there is an urgent need for a framework policy on how to regulate drone proliferation and attacks. The next section is on existing work on drone security and privacy issues.

14 Existing works

Complementary metal-oxide semiconductor (CMOS) chips at the nanoscale enable communication, interaction, and data transmission between aerospace vehicles, relay satellites, 5G terminals, etc. [304]. In extreme radiation settings [202, 306], charge sharing-induced multiple-node-upsets (MNU) have increased the sensitivity of storage circuits in CMOS technology [202, 306]. Nowadays, drones are used in the areas of search and rescue, patrolling, cargo delivery, environmental monitoring, industrial supervision, security, surveillance, policing, and military operations [323]. Drones can also be utilized as wireless relays, flying sensors, and even aerial base stations because of their enormous coverage and ease of installation [103, 166, 218]. Concerns about the commercialization of drones include the possibility of attackers intercepting and controlling a lawful drone's transmissions [88, 126]. Terrorists intercepted real-time video footage from US surveillance drones flying over Iraq in 2009 using Russian software designed to grab satellite television signals. The base stations of US Predator and Reaper drones were infected with keylogging computer malware in 2011. While there is a significant technological gap between capturing video feeds or infecting a computer with a virus and controlling a drone's command link, there is a possibility that attackers will obtain this capability. This threat may be minimal because the US drone fleet's Predators and Reapers' command links are encrypted [246, 251]. Commercial drones' unencrypted command links, on the other hand, might be easily exploited by attackers. The study by Shephard et al. [261] also demonstrated that global

navigation satellite system (GNSS) spoofing may be used to kidnap drones. Drone hacking and hijacking are serious concerns to consider as the number of commercial drones grows [50, 156]. In the aspect of communication link, Guo et al. [105] present a directed network link prediction approach based on local indices path extension because current link prediction indices do not completely exploit the network path.

A sensor and hardware attack on the drone, as well as a network attack, are two ways to deactivate it. Drone sensor vulnerabilities are used in sensor and hardware attacks to deactivate the drone. Sensor assaults in drone systems often involve communication link jamming and GPS spoofing. The communication link between the drone and the GCS or the controller is jammed, making it impossible for the drone's control message to be relayed.

GPS spoofing is a strategy that takes advantage of the vulnerability of the communication between the GPS satellite and the drone GPS sensor, as depicted in Fig. 3. By sending a bogus GPS signal, a GPS spoofing attack is utilized to fool the drone [76, 110]. The GPS signal power can be decreased because the distance between the satellite and the sensor is great in a real GPS signal. By creating GPS signals near the drone, it is feasible to send bogus GPS information to the drone. Tippenhauer et al. [276] investigate a GPS receiver spoofing attack. These attacks either necessitate the use of specific equipment or have a limited attack range; therefore, a method can be used everywhere.

Roddayetal et al. [234] performed an experiment utilizing a man-in-the-middle attack to deactivate a drone. The authors use the Zigbee API mode in their system, which allows them to send broadcast packets to drone networks. The initial vector values are collected by the broadcast packets and utilized to crack the WEP. Rani et al. [232] exploited the Wired Equivalency Privacy (WEP) weakness to attack the drone. This attack approach can only be used against a single drone manufacturer.

Kwon et al. [153] offer a method for hijacking a drone by exploiting a weakness in the MAVLink protocol. It is important to input the NetID to connect to the drone when applying the telemetry module to operate the drone via MAVLink. As a result, if the NetID is known, it is simple to take control of the drone. Highnam et al. [117] take advantage of this by continuously sending malicious MAVLink packets through the use of an antenna with the same NetID. Therefore, there is a need for another method that will not require any additional information, such as a NetID.

Pleban et al. [224] and Samland et al. [242] used the vulnerability of the AR drone to hijack a drone. Samland team employed port scanning of the FTP port and then, sent a malicious code to the drone to gain unrestricted access to the drone's private images and data. Pleban et al.

[224] also used a telnet port vulnerability in the ARdrone to re-install the shell script and restart the AR drone. Pleban et al. were able to effortlessly take control of the AR drone in this manner.

Rani et al. [232] investigated how to deactivate a drone in Wi-Fi networks by assaulting the access point. The authors use the weakness of wired equivalent privacy (WEP), one of the WiFi security protocols, in this study. WEP encryption includes a flaw that allows a specific amount of data to be collected and used to crack the pre-shared key. It is particularly easy to hack the pre-shared key value in WEP encryption with the password cracking program aircrack-ng. The authors use aircrack-ng to deactivate the drone by sending de-authentication packets to it. This attack is limited to drones that communicate via Wi-Fi, but our approach may be used on any drone system that communicates via the MAVLink protocol.

Drones could be used to listen in on electronic conversations. At the Black Hat forum held in Singapore in March 2014, SensePost unveiled their Snoopy drone, which can collect data from unknowing smartphone users. The drone uses the company's software, which is installed on a computer attached to the drone. Without the user's permission, the code might be used to get into devices and steal personal information. This method could be especially effective in a crowded environment where several people's phones are automatically searching for WiFi networks [99, 101]. In February 2015, AdNear, a Singapore-based marketing firm, was discovered to be using drones to acquire signal strengths and other wireless data from bystanders to determine the whereabouts and motions of mobile phones. AdNear may then utilize this information to provide hypertargeted ads and other promotions to users walking past a storefront in the hopes of getting them to enter their marketing firm [246].

Most pilots are concerned about the development of commercially available small drones. A drone may, for example, enter an airplane's engine and shut it off. Furthermore, as drones develop in size and sophistication, the likelihood of mid-air collisions increases, as do the consequences [31, 317]. Helicopter pilots have reported near-misses between commercial passenger planes and drones [173], and there have been instances of near-misses involving drones and commercial passenger planes [101, 273]. In July 2015, it was revealed that hobbyist drones flown into forest fire zones, ostensibly to capture pictures of the fires, forced firefighting helicopters to be suspended until the drones could be retrieved [128]. Officials say hobbyist drones seeking close-up video are endangering low-flying aircraft used to fight wildfires in California, such as an air tanker, that was involved in a blaze near Phelan [128].

The backbone of modern industrial automation is Cyber-Physical Systems (CPS). That is, they are data-driven, unmanned machines. CPS uses wireless sensor computing networks and technologies to connect physical and electronic items. The Internet of Drones (IoD) is thus an example of a CPS. CPS adoption is linked to the notion of "Industry 4.0," which describes the process of combining technology and information to provide autonomy, reliability, systems, and control without human intervention. Unauthorized physical interference with the CPS's work is possible, including collision attacks, jamming; tampering with package contents; denial-of-sleep (DoSL), and cyber-attacks, such as: data manipulation; impersonation of another person's identity.

15 Methodology

Scopus and journal citation report (JCR) was used to do our search, which consisted of a keyword search for 'drone security*'. Thus, we searched for synonyms such as 'drone', 'drone*', 'unmanned aerial vehicle*', 'UAV,' 'unmanned aircraft*', 'unmanned aviation,' or 'remotely piloted aircraft*'. The findings were then narrowed down to the Scopus-assigned subject areas of 'security, privacy, and drone cyber-attack' which cover a wide range of topics such as security, cyber-attack, drone cybercriminals, and commercial or civilian, and military drones. More also, we limited the search to articles such as those published or in the press, conference papers, letters, magazines, or book chapters, and we only looked for publications in the last twenty years, that is, from January 2012 until April 2022. Furthermore, the search only considered English language articles.

Other articles were eliminated because they were superfluous, such as other usages of the word 'drone(s)' for instance, those articles that did not discuss drone's security or drone's privacy. Articles that did not have the full text were also removed. Covidence was used to review the substance of the articles, yielding a final sample of 250 articles. After that, the data were analyzed using Excel and Bib excel [220]. This study employed a total of 250 articles, which covered a wide spectrum of commercial industries as well as civilian applications. It also encompasses changes in cyber-attack behavior and security practices, ethical issues, policy-related security and privacy issues. The purpose of this research was to learn how to recognize drone cybersecurity flaws and cyber-attack issues, analyze them, and provide solutions based on best practices found in the literature. It delivers the following answers to the research questions: What impact does a drone cyber-attack have? What impact does drone cybersecurity have? What are the most effective cybersecurity

strategies that organizations or government may employ to combat drone attacks? How can an individual, an organization or government deal with a drone cybersecurity issue while on a mission?

The types of drone attacks considered by this research team include communication attacks, sensor attacks, hardware attacks, software attacks and physical attacks on the drone itself. Mathematically, the set of risk sub-scenarios that answer these questions are as follows:

$$R = \{ \langle S_\alpha, L_\alpha, X_\alpha \rangle \}, \alpha \in A, \quad (1)$$

Equation (1) is set of risk sub-scenarios, where S denotes the risk sub-scenario, L the likelihood, and X the damage vector or resulting consequences. The index α ranges over a non-definable set, A , that includes all risk scenarios as found in the literature [135].

The goal of this study was to examine drone cybersecurity issues and developing physical security threats based on a scoping review of relevant literature published on the subject. Between January 2012 and April 2022, published articles in journals, letters, newspapers, and magazines. A scoping review, as opposed to a systematic review, is seen to be superior for quickly covering broader topics from publications with a variety of study designs without judging the quality of the papers included.

15.1 Identifying relevant articles

Two complementary search strategies were utilized to find relevant published articles, that is, from January 2012 to April 2022, a state-of-the-art examination of reports was conducted. Search engines: The table and timeline were created using a variety of search engines. These search engines include Google, Yahoo, Baidu, Qwant, Bing, AOL, Excite, Ask.com, DuckDuckGo, etc. More also, we explored search engines for this study, such as Google Scholar, Refseek, Microsoft Academic, Worldwide Science, Wolfram Alpha, and Educational Resources Information Center.

Keywords deployed: When compiling the drone cyber-attack results, a number of keywords were applied. The Google Translate tool was utilized to translate non-English phrases (Google Translate, 2020). Similarly, independent sources were used to validate the translation [267]. The requirement for locating results has been established, and they are presented as comparable to existing evaluations in the cybersecurity as found in the literature. Therefore, the following key terms were used in drone's cyber-attacks.

15.2 Screening and selection procedures for published articles on drone security.

Firstly, an English search was conducted using the Eureka database, which specializes in print media, using the following keywords: "Drone cyber-attacks," OR "unmanned aerial vehicles (UAVs) and Cybersecurity" "Drone and Cybercrime" OR "UAVs and Cybersecurity," "UAVs and Cyber-attacks,". Secondly, using the same terms, a Google News search was run to find other relevant articles. The first search results returned 7577 potentially relevant articles; however, 5013 of them were irrelevant, while 2564 were included. The articles were examined again, and 2061 duplicates were found. The methodical screening was the third step. As a result of the methodological screening, 253 articles were found to be irrelevant for the study, and 250 articles were found to be relevant for the study, as shown in supplementary Fig. 5.

15.3 Selecting relevant articles

The exclusion and inclusion requirements were generated to help in removing articles that were uncorrelated with the core study challenge of analyzing the best practices used by organizations to manage drone's security and privacy issues. Research described corporate policies implemented expressly to address the drone security problems, with a focus on North America, South America, Europe, Australia, Africa, and Asia. Articles that only discussed business or that examined perspectives on drone management without addressing security and privacy issues were omitted. Finally, 7327 articles were excluded throughout the selection process, while 250 relevant ones were included out of the 7577 searched.

15.4 Selection criteria

The articles discussed in this paper are all mentioned in the 'References' section. Meanwhile, exclusion and inclusion criteria on reviewed articles are described in supplementary Table 8.

15.5 Data visualization

We analyzed and filtered these papers using Covidence, which is an online platform that facilitates the speedier review of articles through a workflow of the review process and collaborative review. Drones are the major issue of the study, such as strategies for propagating technology items into the construction sector, according to the preliminary screening results [249]. Although drones were more

important, however, several papers focused on operations, for instance, Zhou et al. [324], or engineering, for example, Chen et al. [56], with little regard for security. The data were analyzed, and a data extraction grid was created using Microsoft Excel. The sheet was divided into six sections: (i) general characteristics, (ii) design and (iii) drone cyber-attacks impact, (iv) organizational techniques for drone cybersecurity management practices, (v) some individual techniques for drone cybersecurity management practices, and (iv) drone cyber-attack preventive measures. More also, to develop the first draft of the list of drone's cyber-attack types, several articles were analyzed.

15.6 Questionnaires and participants

Two sets of questionnaires were designed: one for the general public and one for the experts in the field of aviation. The questionnaires designed for the general public were two hundred, while those for the experts in the field of aviation were one hundred. Two hundred persons were contacted online to assess their understanding of cybersecurity and drone cyber-attacks, which were communicated to the general public, and the results obtained are presented in supplementary Table 10. Meanwhile, one hundred persons were contacted online to assess their understanding of cybersecurity and drone cyber-attacks, which was communicated to the experts and stakeholders in the aviation industry, and the result obtained is presented in supplementary Table 11.

15.7 Getting information, processing it, and reporting

Online questionnaires were taken to address the drone cybersecurity issues in many types of attacks. Information on cybersecurity and drone cyber-attacks was gathered from the general public that participated and also from experts and stakeholders in the aviation industry. One hundred and eighty-seven out of two hundred people from the general public responded to the online questionnaire. Then, 91 out of 100 executives from aviation and related organization's executives responded to the online questionnaire planned to prioritize operational capabilities toward addressing the drone cybersecurity issues in many types of attacks. The information gathered from both aviation experts and stakeholders, as well as the general public, was analyzed. In the following sections, figures and tables were developed to reveal the study's primary findings. The percentage of articles discussing a given item under each category was also assessed, and the results are presented in the figures and tables.

16 Result

Results are presented according to findings in the search keywords for articles, literature review, data gathering, questionnaire responses, analysis, expert opinions and interpretation. We start from the number of articles processed in the study.

16.1 Result on the number of articles processed in the study

The papers that have been identified in the search result are a mix of several paper kinds. While some are case studies of practical use, others are devoted to application and management, but they are also thinking about future drone security endeavors. There are a number of studies written from a non-drone perspective that provide helpful insights into drone security in general. In addition to the bibliography result, this study discovered that drone use cases are a worthwhile topic for discussion. Similarly, drone security present issues have stretched past previous ones to include new ones that have not been addressed.

16.2 Result of bibliography

Drone security articles have appeared in many journals, conferences, magazines, letters, theses, books, and particularly case studies of drone cyber-attacks have become prominent in the literature in recent times, as shown in supplementary Table 9.

Thus, supplementary Table 9 highlights the number of articles studied. It is clear that 250 articles have been surveyed after the filtering process and selection. It summarizes the published sources for the 250 peer-reviewed drone publications. What is clear is that during the screening and selection period, a small number of database sites accounted for a considerable amount of publications on drone security. These sites are Science Direct, Emerald insight, IEEE Xplore, Elsevier, ACM Digital Library and Springer Link as presented in Table 9. A vast tail of single publication sources is also visible. Other sorts of articles, however, continue to exist to tackle specialized drone topics (e.g., application and management). The article mapping gives an overview of the topic's publications, notably in terms of drone cybersecurity challenges.

16.3 Result based on the percentages of excluded and included articles

In this study, 7577 articles were searched and 7327 were excluded throughout the selection process, while 250 selected relevant articles were included accounting for 6%

of the included articles and 94% excluded articles discovered as shown in supplementary Fig. 6.

Supplementary Fig. 6 highlights the number of exclusive and inclusive articles studied as a percentage. It showed that many articles were removed because they were not directly relevant to the study, though they might be related. However, such related articles did not meet the needs of the topic discussed in this study. Hence, they were eliminated without bias to provide accuracy in the study and to allow the researcher to address the problem and find a solution.

16.4 Result based on the types of drone cybersecurity questionnaires given to participants

The drone cybersecurity questions for participants were designed to raise awareness of cyber threats. In today's world, everyone needs to be aware of the cybersecurity threats and risks. This study focused on drone cybersecurity, and specifically on cyber-attack and how they can be addressed. Presently, the majority of people are not informed of this cyber issue, and many of them are not even aware of the potential cyber risks associated with drones. This is creating a vacuum in terms of detecting and preventing cyber-attacks generally. As a result, the vacuum provides an opportunity for cybercriminals to pursue their malicious goals. Two sets of questionnaires were designed: one for the general public and one for the experts and stakeholders in the aviation industry. The survey questionnaire was conducted to assess participant understanding of cybersecurity and drone cyber-attacks. The results obtained are presented in supplementary Table 10.

Given the general public and experts in the field survey results in supplementary Table 10, it is evident that the majority of the respondents are not aware of these issues, making it clear that the public needs awareness on drone cyber insecurity and attack prevention. Manufacturers of drones and researchers need to find solutions to the highlighted problems in the questions in supplementary Table 11. The government should be ready to enact legislation to address drone cybersecurity issues, particularly in terms of proliferation and privacy.

16.5 Result on types of common non-cyber-attacks on drone

Participants response to the questions on awareness of drone's non-cyber-attacks in recent times is presented in supplementary Table 11.

Participants responses to the questions on awareness of drone's non-cyber-attacks in recent-times demonstrate that

majority of people are still not informed of drone security flaws and the consequence.

17 Discussion

The goal of this study was to examine the impact of drone cybersecurity on national security using a scoping review of relevant articles published on this topic in journals, conferences, magazines, media broadcasts and newspapers between January 2012 and April 2022. The study's analysis of the 250 articles provides an overview of the challenges of drone cyber-attacks in global scale. The article contributes significantly to the emerging literature on drone security and countermeasures. It is a review that focuses on both cyber-attacks and non-cyber-attacks in a real-life scenario. According to the results of the survey on drone security, the Software-Defined Radio (SDR) can provide a very versatile platform that can be utilized for a number of Cyber-attack scenarios, reflecting numerous threat vectors that can be launched from a single hardware platform. Commercially available SDR platforms, such as the Hack RF, can pose a threat in three areas of electronic attack: interception, jamming, and packet injection. Drone manufacturer issues: The results of this survey show that drones are vulnerable to a variety of attacks at various levels. Manufacturers' goals of improving the performance of their commercial products over security could be one explanation for this problem.

Another reason is the extra cost that manufacturers must incur in order to include security features. Existing drone manufacturers must focus on security and privacy when creating their products at all stages of the supply chain. Participants' answers to questions about recent drone non-cyber-attacks show that the vast majority of individuals are still unaware of drone security weaknesses and their consequences, as seen in Tables 10 and Table 11. However, the majority of participants indicated that more study is needed to solve drone attack dilemmas, as well as financing to conduct research that will lead to solutions to the aforementioned drone issues. Given the results of the general public and experts in the field survey in Table 11, it is clear that the majority of respondents are unaware of these difficulties, indicating that public awareness of drone cybersecurity and attack defense is required. Drone manufacturers and researchers must provide solutions to the challenges mentioned in Table 11's questions. The government should be ready to pass legislation to address drone cybersecurity issues, including the aspect of proliferation and privacy.

17.1 Comparison of security threats' intensity by different types of cyber-attacks on drones

Drone cyber-attack can cause many damages to economy such as loss of personal information, financial and medical information. These cyber-attacks can also damage the reputation and safety of individuals and companies. In this study, we compared the damage intensity caused by different types of cyber-attacks on drones, which were achieved using twelve references and thirty-one areas of drone cyber-attack techniques. These techniques could be found in many studies, as presented in supplementary Table 12.

Where symbols ✓ represent area of coverage of types of drone cyber-attacks in each study, and ✗ represent area does not cover in the study.

17.2 Comparison of result with some state-of-the-arts surveys on drone security

A comparison of the results with some state-of-the-art surveys was achieved using the nine levels of research focus, such as emerging security issues, communication-based attacks, communication, sensors, hardware, hardware-based attacks, software attacks and physical attacks on the drone itself. A comparison of the results with some state-of-the-art surveys is shown in supplementary Table 13.

Where symbols ✓ represent area addressed in each survey, and - represent area not addressed in the survey.

Thus, supplementary Table 13 revealed that this study contributes to knowledge in many areas compared with other state-of-the-art surveys on drone security.

18 Future directions

There is a substantial possibility of an increase in drone cybercrime in the near future. Since vulnerabilities are linked to drones, the damage opportunity for hackers increases, and cyber attackers will continue to ramp up their mischievous actions and adopt more advanced and sophisticated modi operandi. Hence, we provide intriguing security and privacy research directions for drones in this section, which could be addressed in future studies.

- (i) *Secure Communications for Drones* There is a need for adequate drone communication protocols to ensure reliable and secure connections between different components of the drone system. Similarly, there is a need for research on safeguarding drone

communication routes while obtaining maximum network capacity. Furthermore, drone authentication can protect the communication connection or link by preventing replay and impersonation attacks [235]. The development of drone access controls, such as authorization and authentication mechanisms, remains a difficult research area. Unauthenticated drones must not engage in flight missions or collect data from other drones in the ecosystem. Multi-drone missions, on the other hand, are enabled through the utilization of specific networking models for drones, such as FANETs [34]. FANETs, on the other hand, are vulnerable to a variety of threats [61], and enabling secure communication in a multi-drone network is still a work in progress.

Despite the fact that several FANETs routing protocols have been proposed in the literature, none of them have been implemented (Arafat and Moh 2019), and they appear to be unable to fully meet the security and privacy standards, indicating that more research is needed in this area [175]. Although, co-site interference has grown in importance with the proliferation of communication systems on independent platforms as found in aircraft and ships. However, the nearest matching method can be used for broadband interference [130].

- (ii) *Intrusion Detection Systems for Drones* Detecting intrusions targeting drones during a flight operation necessitates a real-time network traffic analysis. To that purpose, deploying an IDS for drones allows for the detection of several types of intrusions, including signal manipulation, malware, routing assaults, and message forgery [60]. Furthermore, developing anomaly detection frameworks to track harmful actions is critical for spotting attack trends [247]. In addition, using honeypots and honeynets in conjunction with an IDS can assist defend a flight operation from malicious individuals or groups [89]. The intrusion modes, effects, and criticality analysis (IMECA) method was used to assess the invasion of attack in some systems. Since drone networks are a complex cyber-physical ecosystem with several components, intrusion detection methods should take into account various data collection sources to improve performance. More information sources, on the other hand, can raise

communication costs and lead to significant computing overhead. Due to the present tradeoffs between performance and security, implementing such solutions can be difficult. As a result, lightweight IDSs are required to track communications and detect attacks to prevent attacks. Some systems use flight behavioral profiling to detect anomalous activity and malicious intrusions in this regard [43]. Cyber-attacks that compromise drones while maintaining a constant flying pattern, on the other hand, are undetectable.

- (iii) *Implementations that are realistic* Drone development and deployment in practice necessitates a focus on security and performance compromises. It is important to address the drone system's security and privacy requirements from a security standpoint. Furthermore, from a performance standpoint, we must consider energy, computing expenses, and storage overheads. Cryptographic primitives are used, for example, in the implementation of authentication methods and the development of lightweight cryptography algorithms for energy-constrained drones. Such implementations, on the other hand, may consume too much energy and raise the computing cost. Therefore, developing a strategic solution that balances both viewpoints is a big unsolved research question. Existing security countermeasures only work in certain hardware and software configurations. Thus, we must evaluate the potential deployment issues across various drone systems when proposing real-world solutions. Unifying a deployment interface for different types of drone systems could be a feasible option. It should also be noted that simulating cyber-attack scenarios on drones prior to implementation could indicate the resilience of current cybersecurity measures. Furthermore, available simulation platforms for drone security assessments are restricted [69], indicating that this area warrants further investigation.
- (iv) *Protection of Personal Information* The integration of drones into national airspace has sparked concerns about privacy protection. Hence, with these implications, critical data obtained by drones are leaked. Although the acquired data may be uploaded to third-party entities for processing or storage, such as cloud servers, nonetheless, there is a need to

preserve the privacy of outsourced data in this setting. In the existing literature, various private data or information techniques have been presented. Applying privacy-enhancing technologies to maintain consumers' data and ensuring privacy and security with third-party organizations are examples of how to mitigate privacy data invasion threats. Secure computing or divergent privacy mechanisms, for example, will protect individual users' privacy while allowing data coordination among drones while at the same time maintaining privacy. Other instances include using homomorphic encryption to execute computations on encrypted data [8].

- (v) *A strong password and WPA2* Similarly, using WPA2 and a strong password to secure the drone access point is important [288]. To gather useful information without revealing it, researchers used a new hybrid chaotic system with applications in image encryption [15] and zero knowledge proof (ZPF) [141, 150, 279]. Furthermore, any other form of cryptographic or encryption techniques [208, 210–212, 318] can be applied to secure a drone's data while maintaining its privacy. Therefore, urgent research is required to fortify data that are been communicated during drone's flights.
- (vi) *Data Aggregation in a Secure Environment* The widespread use of drones in a variety of applications has increased the amount of data collected and shared. The information collected is usually pooled in order to make the best use of the available resources. However, malevolent actors must be safeguarded from the data collection or aggregation process. When deploying aggregation methods, encryption mechanisms should be considered to ensure confidentiality, that is, to allow for secure data flow between the ground control system (GCS) and drones. Furthermore, developing efficient and secure data aggregation systems for drones would lower energy and communication costs while concurrently maintaining privacy. Nevertheless, developing such strategies is a never-ending task.
- (vii) *Forensics of drones* When a security breach occurs during a flight operation, usually forensics analysts can be called upon to examine the infected drones [1]. However, gathering evidence from drones that do not have logging capability is likely impossible

[22]. Essential information such as onboard-flight statistics and flight trajectories is particularly kept in the flight controller's dynamic random-access memory (RAM), thereby making recovery a difficult procedure. Hence, creating models is critical for providing in-depth drone forensic analysis [17, 18, 21, 30]. Even with robust forensics models, anti-forensics strategies may impede digital investigations [23]. One proposed mitigating technique is to use a forensic-by-design approach, which incorporates forensic needs into the drone system's architecture [5]. Drone forensics is a new area of drone security research that has yet to be investigated.

Current digital forensics approaches are insufficiently unified and standardized to cover a larger range of commercial drones. This is a significant problem in drone forensics, as an attacker could possibly hack specific drones whose forensic models have yet to be addressed. With the rise of cyber terrorism, there is a pressing need to learn how to prevent it using modern scientific approaches such as AI [75], and ML algorithms [321]. Using feature selection optimization [2, 3], feature extraction [322, 325], big data application techniques [210–212], mathematical models [203, 204], forensic DNA profiling [203, 204], and cognitive deception model [207, 272] can also help in securing drones against malicious attackers.

- (viii) *Emerging Technologies* Recently, emerging technologies such as artificial intelligence (AI), software-defined networking, fog computing, and blockchain technology have been extensively used to protect drones [111, 211, 270]. These technologies are used in a variety of civilian settings. Blockchain technology's distributed architecture adds an extra degree of security to communication [100]. With smart contracts, strong cryptographic algorithms to save data as a blockchain, and consensus methods, tampering with drone communication has become more difficult. However, the major applications of blockchain for drone communication security suffer from real-time deployment for highly mobile drones [151]. The principal uses of Blockchain for drone communication security, on the other hand, are hampered by the need for real-time deployment for highly mobile

drones [151]. Furthermore, the application of Blockchain technology to secure drone networks in the real world is still a work in progress.

The advancement of AI technologies, such as ML, has shown enormous benefits for security-oriented applications, such as defending drone networks from cyber-attacks and privacy leakage. Various ML-based security frameworks have been presented in the literature to address various security concerns, such as the detection of DoS and other malicious drone attacks [309].

- (ix) *Federated learning strategies* These strategies have recently been demonstrated to outperform traditional machine learning algorithms. For example, in IoT networks, drone authentication methods based on the drone's Radio Frequency characteristics are being applied [314]. Notwithstanding, there is a scarcity of drone datasets on which to train machine learning algorithms, for instance, network traffic datasets, malware datasets, etc. Additionally, certain machine learning algorithms may be unable to detect cyber-attacks on drones [177]. Drone communications can be made more secure by using secure data network (SDN)-based drone networks. That is, for various security applications, SDN technology provides dynamic flow control and a customizable network. As a result, the drone network will be protected from future cyber-attacks. The significant end-to-end delay for non-delay-tolerant drone applications is a major disadvantage of utilizing such a technology.
- (x) Aside from that, integrating drones into smart cities necessitates the processing and storage of a vast amount of data. Fog computing technology can aid in the processing and storage of data. Furthermore, fog computing also enables salable and low-latency secure communication between the flying drones and the fog layer. However, the existing fog architecture is not designed for drones, and implementing one could increase data processing time, especially for multi-drone networks. Fog computing, also known as fog networking or fogging, is a decentralized computer architecture that exists between the cloud and data-generating devices. This adaptable framework lets users optimize performance by placing resources, such as apps and

the data they generate, in logical spots. In the future, drones will feature a variety of new technologies [194]. Therefore, there is a need for more study from academia, business and military into the use of developing technology to protect drones in civil and military applications.

- (xi) The smart aerial monitoring systems (SAMS & SAMS-T) from Easy Aerial are surveillance and security drones that have been used to monitor crowds with great success. They are one of the best drones for surveillance and security since they are mobile, resilient, and completely autonomous drone-in-a-box options. Furthermore, the drone can be controlled entirely from afar, making the technology stand out in a crowded market. They can, however, be investigated in terms of the security of the hardware and software in order to improve communication, fast camera motion [321], data capture, and sensor efficiency.
- (xii) *Drones can be compromised or used to compromise other electronic devices* A hacker does not even need their own drone; they can hack yours in a variety of ways to make it serve their needs. The cybersecurity issues will only become more pressing as the number of drones in our skies grows and hackers become more adept at detecting flaws in drone security. As a result, more research is urgently needed not only to address the massive number of drones in the sky, but also to find solutions to many security breaches of drones while in flight, as well as their vulnerabilities to hackers.
- (xiii) *Exploring blockchain technology* Blockchain is a new approach to cloud computing that combines point-to-point transmission, consensus, encryption, and distributed data storage. Attackers may find it difficult to change or remove records that may be used to record, gather, and search information about drone systems using blockchain technology. That is, for example, where the drone has been or what it has done, because it is capable of decentralizing and encrypting communication capabilities, particularly data redundancy checks. It could be used to detect unauthorized drone systems or unusual operating modes, making it more difficult for attackers to collect data from drones or change flight plans. Although it is unknown how blockchain will affect

drone security, it will be a future technology and development pattern.

19 Recommendations

The following recommendations, which include ways to regulate, detect, and stop attacks, are being looked at as possible solutions to the drone cybersecurity problem:

19.1 Regulatory strategies

Regulatory agencies such as the United States' Department of Homeland Security (DHS), the Federal Aviation Administration (FAA), and the National Aeronautics and Space Administration (NASA) must collaborate with policymakers to achieve a coherent drone cyber strategy. Likewise, the Unmanned Aircraft Systems Traffic Management (UTM) Agency, the European Union Aviation Safety Agency (EASA), must continue to collaborate with policymakers, cybersecurity professionals, and other government and law enforcement organizations to achieve a coherent drone cyber strategy. This effort will entail inventorying and categorizing drone platforms, evaluating the potential effects of drone-related cyber-attacks as well as mitigation strategies, and remaining up-to-date on new technological advances that may modify the threat landscape. Moreover, understanding people's cyber behavior can help lawmakers make more informed decisions [159].

The regulatory authorities should also prioritize the most serious flaws and devise methods to limit attack channels and protect attack surfaces. To better understand mitigation alternatives, regulators, such as the DHS, NASA, UTM and EASA, will need to keep track of technological advancements in counter-drone systems and experiment with new attack strategies and technologies. A coordinated and updateable system of monitoring and intervention is necessary because the innovation cycle of cyber-attack and countermeasure shows that even hardened systems cannot be proven immune to attack.

Regulatory bodies will need to keep an eye on drone uptake and foresee the consequences of extensive drone use. Drone applications will be expanded due to capabilities such as swarming and autonomous flight. The number of legitimate-use drones in the air at any given time will grow as drones are used in a greater range of activities. One of the most critical issues in this emerging drone-dense environment in terms of attack mitigation will be identifying legal from illegal activity.

Drones could be used by regulatory authorities like the DHS and EASA to surreptitiously observe targets or suspected opponents; for example, drones could acquire

access to a local network suspicious of coordinating illegal activities such as smuggling operations. This type of access could also be used in overtly aggressive ways, such as deactivating or feeding false information to connected security cameras and alarms right before a regulator raid.

Offensive cyber operations involving drones could also be useful. When ground or manned platforms lack the mobility or quick response required to pursue these transient targets, regulatory bodies may use drones to catch or disable cyber-criminals' drones involved in smuggling. In a disaster zone, DHS may use a drone to jam or deactivate criminal drones, allowing clear passage for helicopters. A more targeted variant might feature a regulatory drone attacking just drones that do not transmit an identifiable friend-or-foe signal.

Policymakers and purchasing authority must ensure that the regulator's purchases of drones address security concerns as a result of the research and industry participation. The Management Directorate and the Office of Strategy, Policy, and Plans both have a say in departmental policy. All components and departments that will use drones must support the drone security policy and establish incentives for the acquisition of secure drones.

Finally, regulators such as the DHS and EASA may deploy their own "guard dog" drone unit to patrol sensitive physical areas and destroy any adversary drone discovered near any drone component under threat of infiltration or data retrieval by an enemy drone. This feature may be particularly appealing to CISA, which deals with essential infrastructure, and the US Secret Service, which deals with federal special security events. Due to wireless networks and future IoT concerns, practically all security settings, including regulatory agencies and components, could be potential targets.

19.2 Detection and prevention strategies

- (i) Detecting drone cyber-attacks with artificial intelligence: Some developing technological developments may be able to help lessen the dangers connected with intrusion detection and tampering with system operation that were discussed previously in this chapter and other sections of the study. One such development is the use of artificial intelligence (AI) techniques such as machine learning (ML) and deep learning (DL) to identify cyber-attacks in real time. The intrusion detection problem in cybersecurity entails detecting harmful activity and rule violations among those who use a cyber system. ML and DL approaches for network and system communications have been proposed by a number of researchers [46, 49, 149, 164, 165, 295, 313]. For

instance, Buczak and Guven, in a 2016 survey paper on the subject, noted that methods that have been applied include neural networks, decision trees, support-vector machines, ensemble learning, and others. These are all common ML methods. By monitoring network traffic, these techniques can detect various types of intrusions and attacks. The authors find that the "huge gap" preventing more successful applications of such techniques is the absence of training data, that is, datasets that contain known intrusions and could be utilized to train algorithms [39].

- (ii) Passive control mechanisms: more research is critically needed on passive control mechanisms for the drone's RFCV. Also, more research needs to be done on a subsystem that checks the physical parameters of a transponder and the electromagnetic environment at the time.
- (iii) Developing cyber-protected wireless systems: Further study should be conducted in the direction of approaches for developing cyber-protected wireless systems with the Wireless 2.0 architecture [92], based on 5G network technologies and intelligent self-control. Intelligent Radio Signal Processing and Cognitive Radio are two examples [40, 114, 221]. Researchers are looking into Mobile Edge Computing (MEC), AI algorithms, Machine Learning algorithms, physically unclonable functions (PUF), block chain, and Elliptic-Curve Cryptography (ECC) to make sure that data communication in drones is safe.
- (iv) It is critical to prevent software and hardware vulnerabilities by sending network packets between the drone and app over a secure channel and upgrading the software running on the devices.
- (v) Online public opinion crisis model can be created to report many incidences of drone's attack such as using a multi-level index system [183].
- (vi) Regulatory bodies can use three types of mitigation methods to better position themselves in light of current and new trends. These are:
 - (a) Regulatory bodies such as the NASA, FAA, UTM, DHS and EASA should cooperate with the drone manufacturing business.
 - (b) Drone makers, policymakers, and other people with a stake in the industry can create a setting where drones can be used safely and with the least amount of risk.
 - (c) All regulatory body components and offices can decrease the risks of attacks by securing and hardening their networks. Then,

developing a defensive scheme to randomize drone implementation configurations, using secondary data sources to detect network tampering, and performing functional integrity checks in the field for drones as vector.

20 Conclusion

This is a general literature review with an informative purpose, which might suggest that there is a possibility of a subjective selection of literature. Notwithstanding, the most referenced publications were found in databases such as IEEE Xplore, Taylor and Francis, Springer, Emerald Insight, Elsevier, Wiley, the ACM Digital Library, Google Scholar, Semantic Scholar, and EBSCO. We conducted an extensive assessment of drone security and privacy issues in this study. Drone security vulnerabilities were thoroughly investigated at five levels: the communication level, the sensor level, the hardware level, the software level, and the privacy level. Likewise, we discussed drone security and privacy, as well as emerging security issues of flying drones, such as other non-cybersecurity challenges of drones like terrorism, mid-air collisions, illegal surveillance, smuggling, electronic snooping, and reconnaissance, then offered potential solutions and future research topics. Next, we presented the experience from the security and privacy aspects of drones, and also provided possible future research perspectives.

Due to the increasing prevalence of commercial drones operating in civilian airspace, issues of national security, including those related to privacy, have risen to the forefront of public attention. Therefore, organizations from different sectors (including business, education, industry, the military, and law enforcement) need to collaborate to develop new security frameworks, standards, and rules. The next generation of commercial drones is being released to the market by existing manufacturers right now, but security and privacy concerns are still lagging behind. However, because of the importance of drone applications, security has become a key problem, necessitating the development of effective solutions to safely guide and preserve stolen data from hackers and destructive acts by criminals. This survey is a useful resource for both manufacturers, and researchers who want to learn more about constructing and designing secure drone designs. Research gaps in the study will provide research focus on how to address the security issues with drones. The major limitation of this research was the lack of co-operation from some stakeholders, like the manufacturers and regulatory

organizations, to state out their problems in addressing the issue of flying drone cyber-attack detection and prevention.

DDoS attacks are predicted to increase in volume and frequency due to the growth of the Internet of Drones (IoD) technological developments, the demand for access to sophisticated systems, and historical trends. Cyber thieves are likely to continue looking for and exploiting weaknesses in these systems in order to use them in DDoS and RDoS attacks. Plans to deploy 5G capabilities, together with recent developments and the ongoing digital transformation, have created new opportunities for threat actors to infiltrate. Monitoring pre-existing and new technologies, ensuring new regulations are followed, and adhering to security best practices for managing enterprise networks and remote workforces are all key aspects of securing critical infrastructure. Every internet user has to understand the basics of cyber hazards and risks.

The contribution of this paper should, however, be considered in light of some other limitations. A longitudinal study should be the focus of future research to uncover these problems and potential solutions. Likewise, future research will focus on drone communication issues and cyber-attacks. The survey covered publications published between January 2012 and April 2022.

Drone technology has sparked a surge in commercial interest that shows no signs of abating. Drones and drone technology, in reality, appear to have a bright and exciting future. Drones in future generations are expected to have increased autonomy as well as greater safety and regulatory norms. These and other improvements have the potential to have a significant impact on a variety of industries, including commercial transportation, logistics, and the military. On the other hand, the security and privacy aspects of drones require urgent attention from academia, manufacturers, aviation, the military, and the government.

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Declarations

Conflict of interest The authors declare that they have no competing interests.

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